



TARGET-90

JKBOSE CLASS-11TH

Biological Classification

One shot

BIOLOGY

ARSHID Sir



Topics To Be Covered

1 Complete Chapter.

2 ...

Introduction



Biological Classification: process by which living organisms are grouped into convenient categories based on some easily observable characters.

Purpose: To make study of living organisms easy and convenient.

Aristotle:

- First to attempt a more scientific basis for classification.
- Used simple morphological characters (eg. Habit) to classify plants into:
 - a) Herbs ✓
 - b) Shrubs ✓
 - c) Trees ✓
- He divided animals into:
 - a) Enaima (possess red blood)
 - b) Anaima (does not possess red blood)

Kingdom Classification Of Living Organisms

1. Two kingdom classification. } *Syllabus*
2. Three kingdom classification. } 
3. Four kingdom classification. }
4. Five kingdom classification. } *Syllabus*
5. Six kingdom classification (3-Domain classification) } *Syllabus*

Two Kingdom Classification

- Introduced by C. Linnaeus

- Living organisms — divided into 02 Kingdoms:

1) Kingdom Plantae: ^{Chlorella (unicellular, Non-motile)} ^{Spizogya (multicellular)} Algae, Bryophyta, Pteridophyta, Gymnosperms,
Angiosperms, ^{Heterotrophic} Fungi, ^{Prokaryote} Bacteria, Euglena

2) Kingdom Animalia: Mammals, Birds, Fishes, Protozoans

Criteria followed:

- Cell wall (present in Plants, absent in animals)
- Mode of Nutrition
- Locomotion

Two Kingdom Classification

Drawbacks:

1. Unicellular & multicellular organisms — placed together
2. Prokaryotes & Eukaryotes — placed together
3. Autotrophs & heterotrophs — placed together
4. Position of some organisms — was controversial

Ex. Chlamydomonas → Cell wall + nt (plant like)
 ↓
 Unicellular motile algae
 → Motile — (Animal like).
 (Flagella + nt)

Two Kingdom Classification

Introduced by Ernst Haeckel.

All living organisms

- 1) Kingdom Plantae
- 2) Kingdom Protista
- 3) Kingdom Animalia

(2) Kingdom Protista (Unicellular Eukaryotes):

- Chlymadomonos
- Protozoans (Amoeba + Paramoecium)

Four Kingdom Classification

Introduced by Copeland.

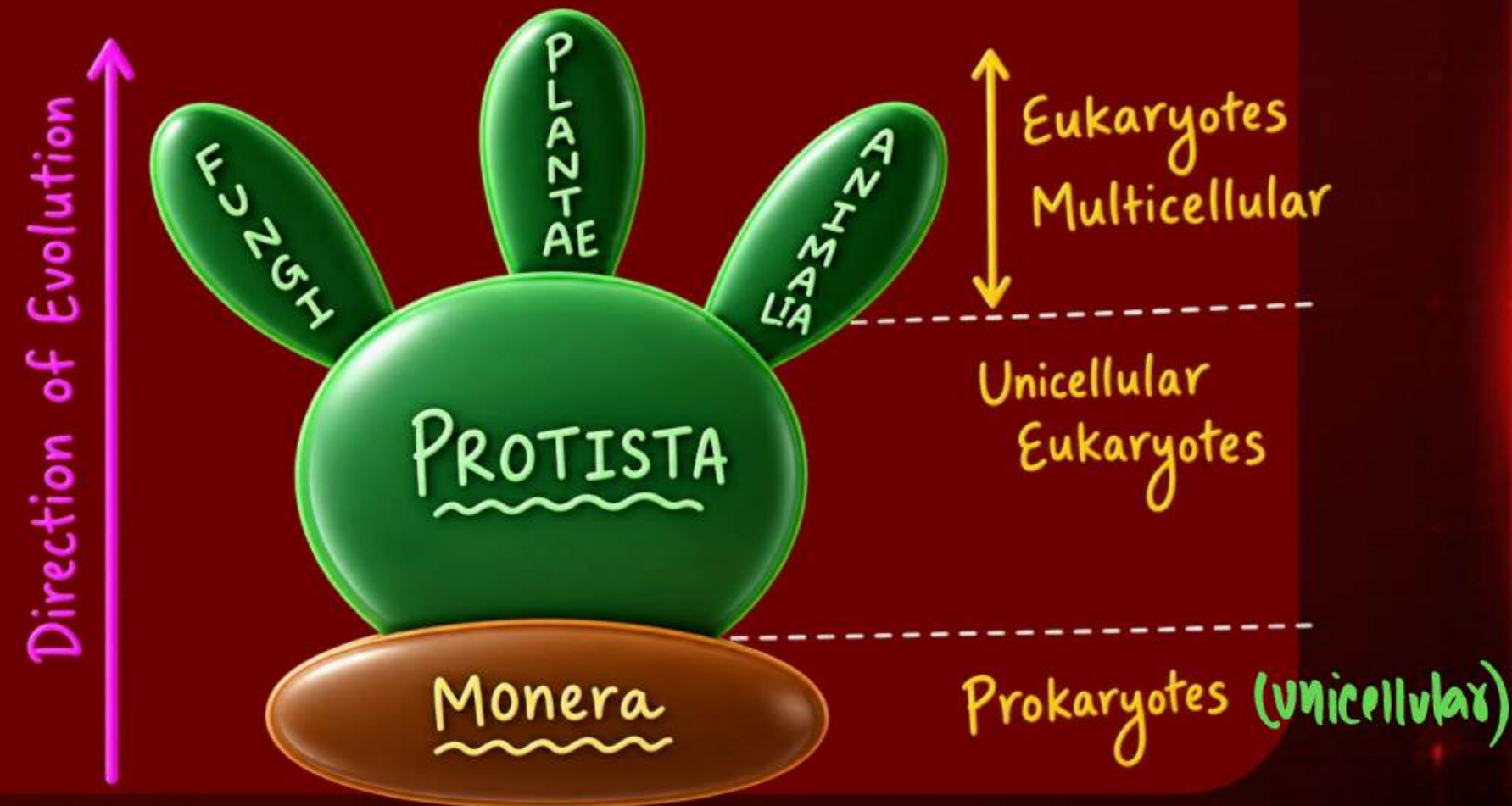
All living organisms

- 1) Kingdom Plantae
- 2) Kingdom Protista
- 3) Kingdom Monera (Bacteria)
- 4) Kingdom Animalia

Five Kingdom Classification

- Introduced by R.H Whittaker (1969).
- All living organisms** – divided into 5 kingdoms.

- 1) Monera ✓
- 2) Protista ✓
- 3) Fungi ✓
- 4) Plantae ✓
- 5) Animalia ✓



Five Kingdom Classification

Criteria followed: (imp.)

1. Complexity of cell str. (Prokaryotic / Eukaryotic)
2. Body organization (Unicellular / Multicellular)
3. Mode of Nutrition (Autotrophs / Heterotrophs)
4. Mode of reproduction
5. Evolutionary or phylogenetic relationship among organisms

Drawbacks:

1. No place for virus, viroids & prions
2. No place for Lichens.

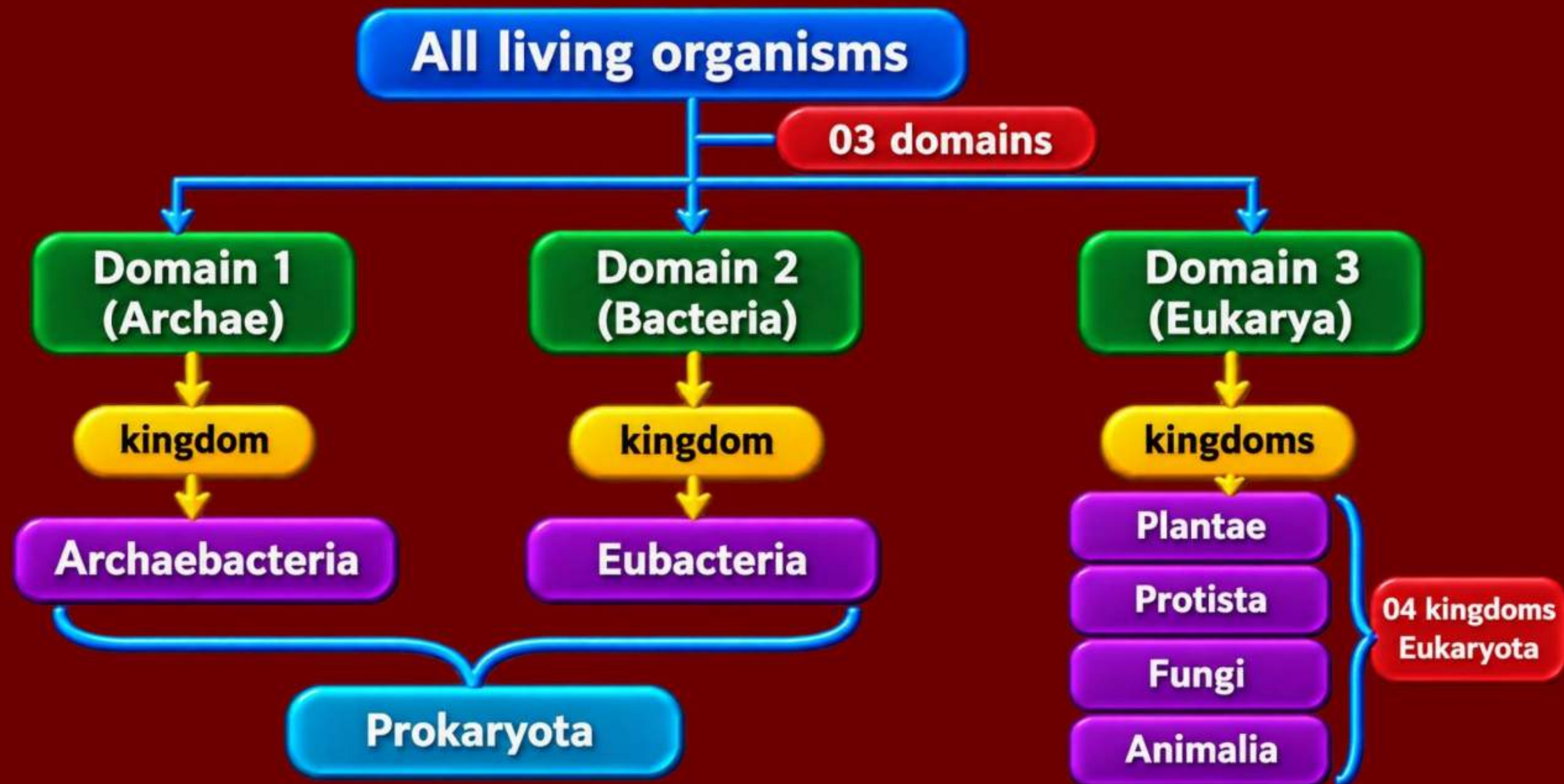
Six Kingdom Classification



Domain = Extra taxonomic category.

(3-Domain Classification)

- Introduced by Carl Woese.



Six Kingdom Classification

Domain → Super Kingdom Category

- 1) Domain ✓
- 2) Kingdom
- 3) Phylum / Division
- 4) Class
- 5) Order
- 6) Family
- 7) Genus
- 8) Species



Kingdom Monera



Bacteria — Sole (only) members.

Kingdom of Prokaryotes.

HABITAT:

- Cosmopolitan (Everywhere)
- A handful of soil contains hundreds of bacteria.
- Snow, soil, water, air, etc.

Kingdom Monera

SHAPE: Variable

- 04 types of bacteria — on the basis of shape:

1) **Coccus:**

- ✓ Spherical 
- ✓ Smallest.

Eg. Streptococcus pneumoniae

2) **Bacillus:**

- ✓ Rod shaped
- ✓ Eg. E. coli



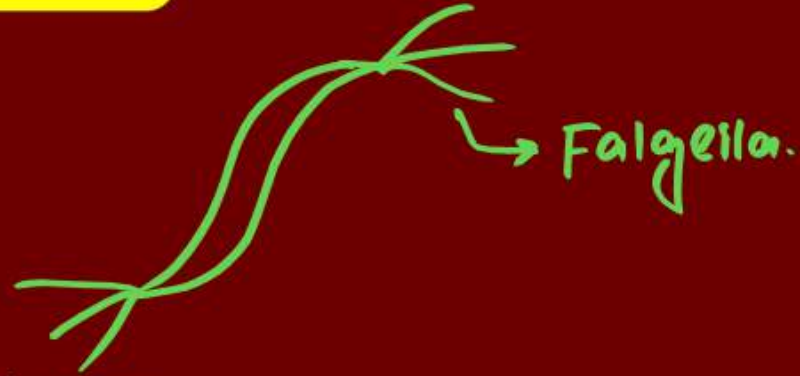
Kingdom Monera



3) Spirillum:

- ✓ Spiral shaped

Eg. Spirillum volutans



4) Vibrio:

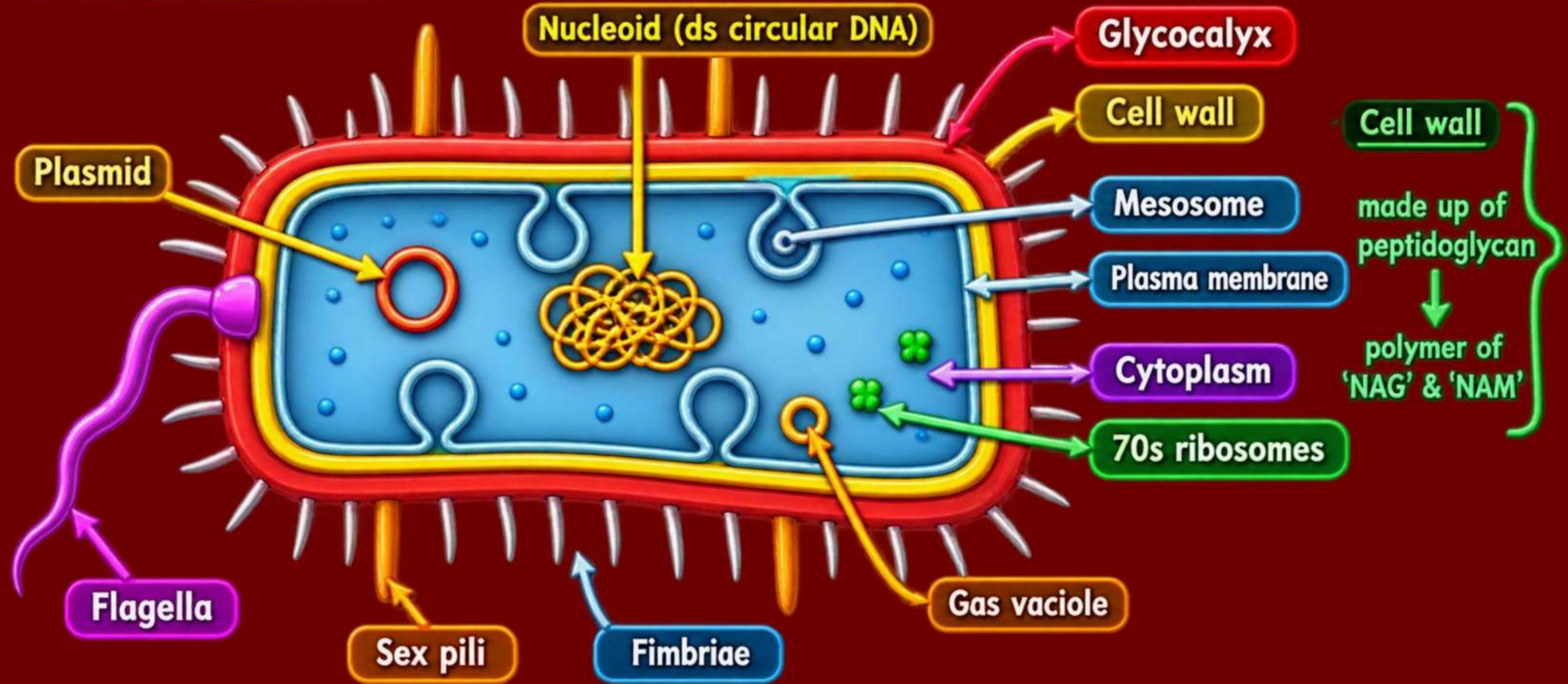
- ✓ Comma shaped

Eg. Vibrio cholerae



CELL STRUCTURE:

typical prokaryotic cell structure



Nutrition in Bacteria

Simple in structure but complex in behaviour.
Shows most extensive metabolic diversity.

Nutrition:

1. Autotrophic

- a. Photosynthetic Autotrophs ✓
- b. Chemosynthetic Autotrophs ✓

2. Heterotrophic

- a. Parasitic ✓
- b. Saprophytic ✓
- c. Symbiotic ✓

Note:

- 1. Heterotrophic Bac. > Autotrophic Bac.
- 2. Among Heterotrophs, Saprophytic bac. are maximum.

Autotrophic Nutrition



1. Autotrophic:

Light/chemical energy is used.

a. Photosynthetic autotrophs:

- Light energy is used.
- Photosynthetic pigments present.

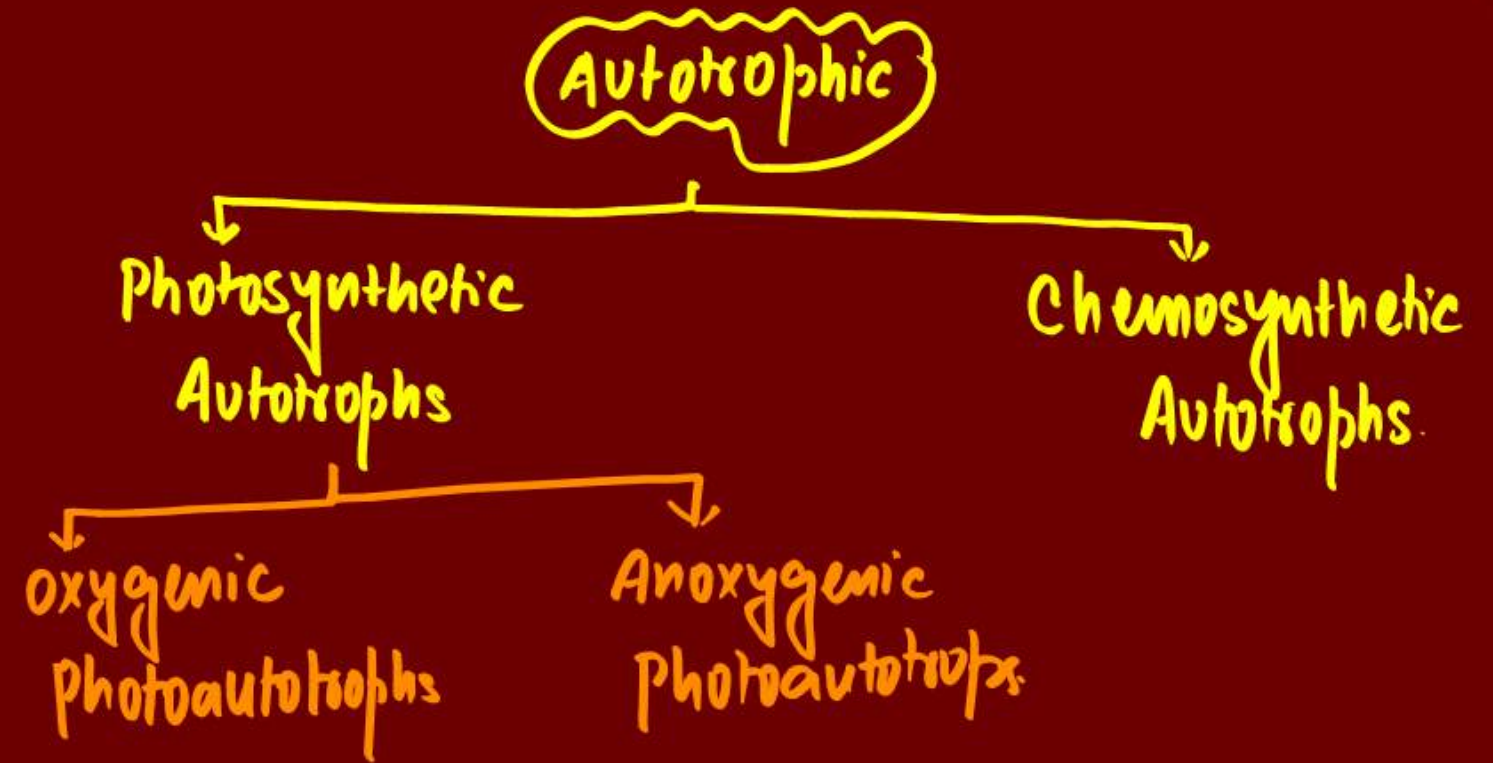
Two types:

i) Oxygenic photoautotrophs

By-product is oxygen.

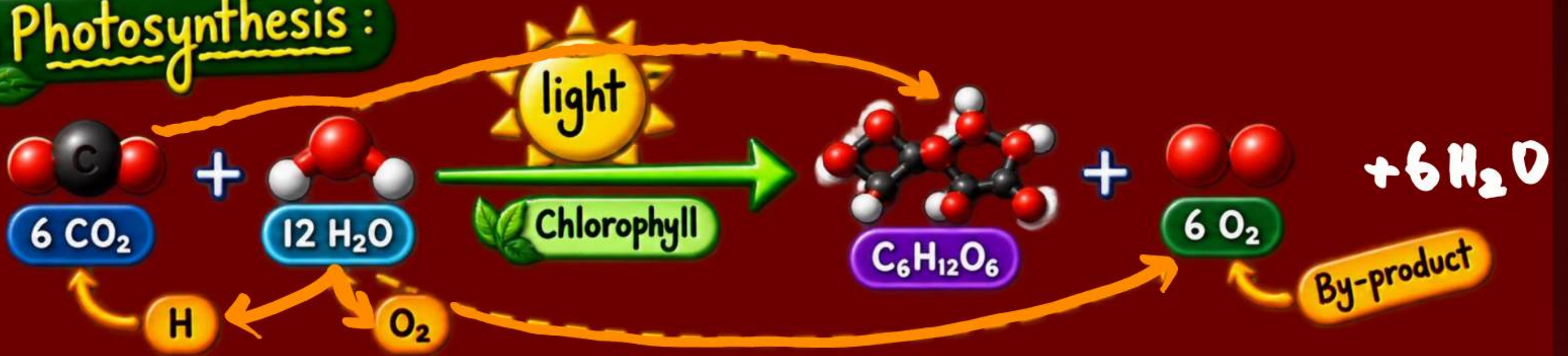
H₂O is H-donor.

Ex. Cyanobacteria ✓

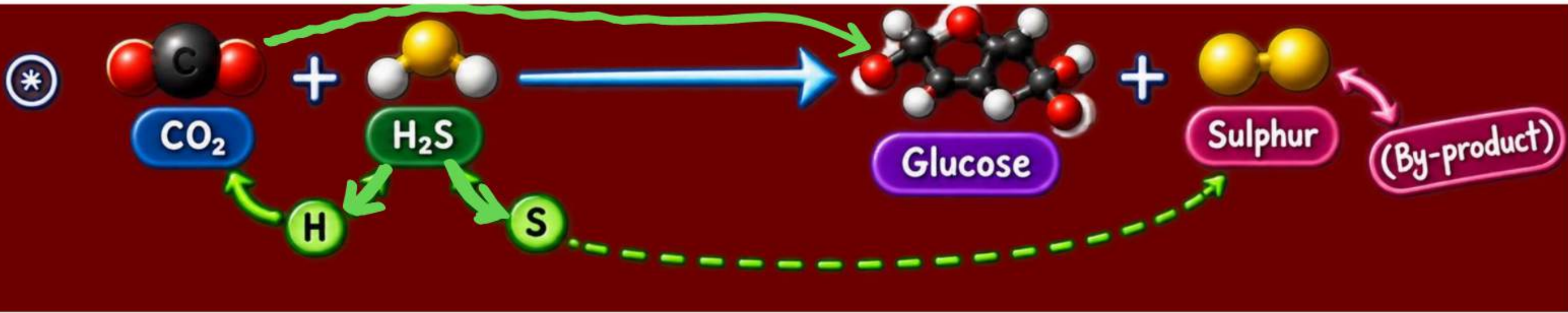


Oxygenic photosynthesis

Photosynthesis:



Anoxygenic



Autotrophic Nutrition



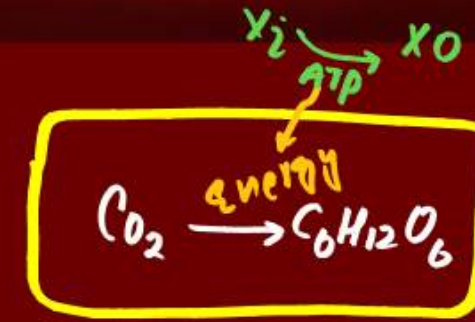
ii) Anoxygenic photoautotrophs

By-product is other than O_2 (sulphur).

H-donor is other than H_2O (H_2S)

Ex. Purple Sulphur Bacteria, Green Sulphur Bacteria

Chemosynthetic Autotrophs



b. Chemosynthetic autotrophs: $\subseteq \rightarrow$ which

Chemical energy is used. C is obtained from oxidation chemical compounds.

Ex. Nitrosomonas, Nitrococcus, Nitrobacter.

Note: Play an important role in nutrient cycling e.g. N, P, K, S etc.

2. Heterotrophic:

- Obtain their food from dead organic matter or living organisms.
- Following types:

a. Saprophytic

- Source of food is dead organic matter.

Ex. Lactobacillus. ✓

b. Parasitic

- Obtain food from other living organisms (host).

Ex. Xanthomonos, Agrobacterium.

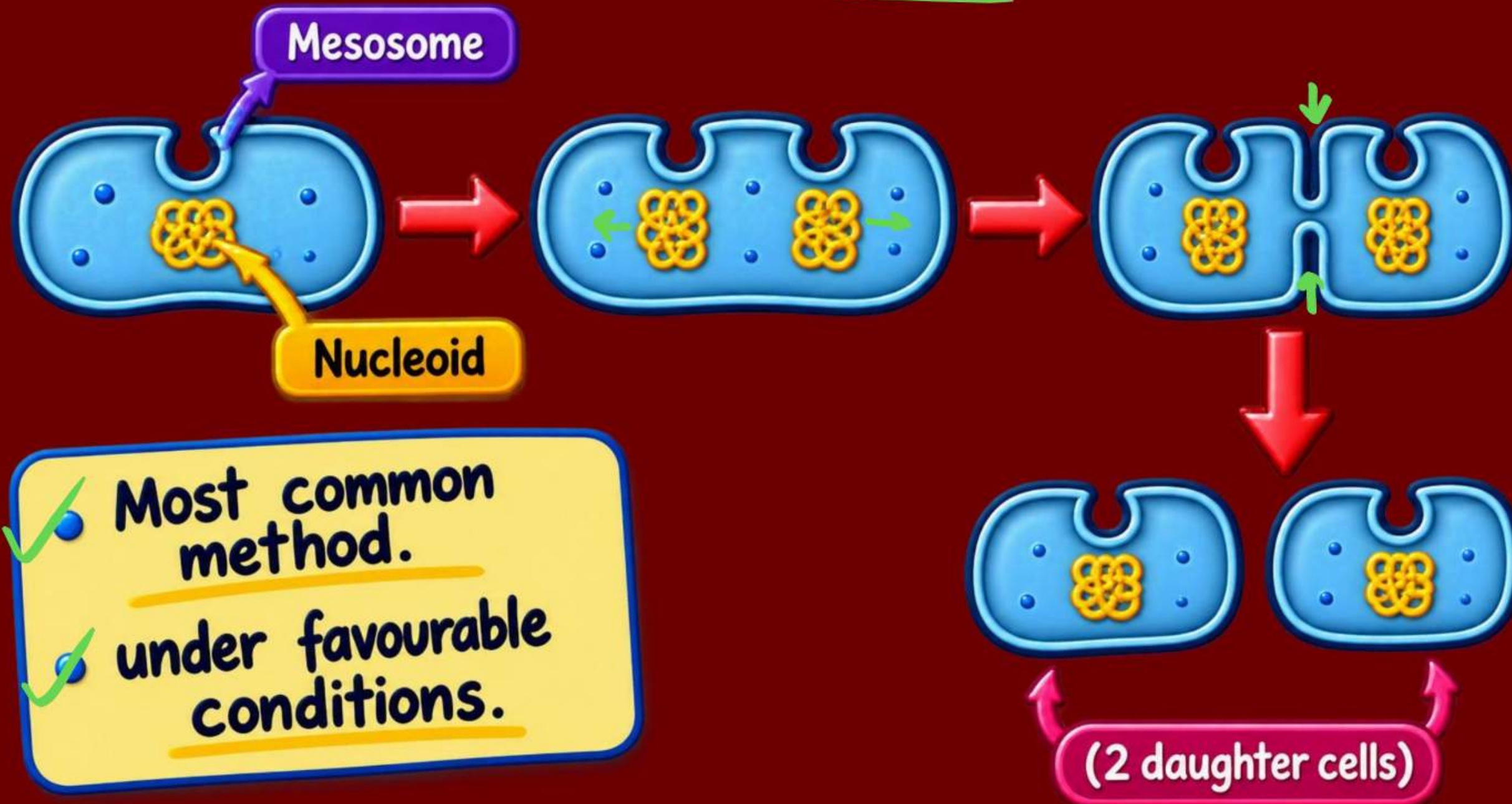
c. Symbiotic

- These bacteria convert atmospheric Nitrogen into nitrogenous compounds like amino acid, salts of ammonia etc.

Ex. Rhizobium

Reproduction in Bacteria — Vegetative & Asexual

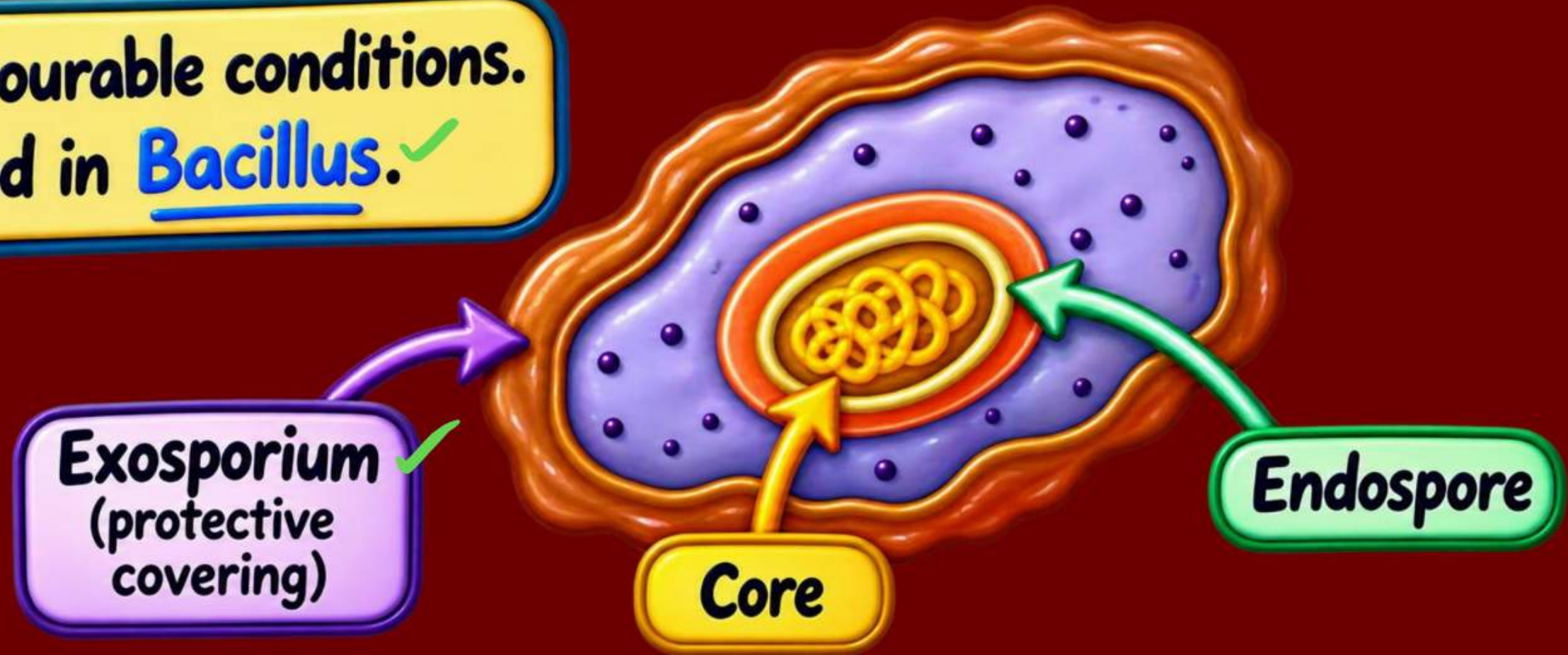
1. Vegetative reproduction — By Binary fission



Reproduction in Bacteria — Vegetative & Asexual

2. Asexual reproduction — By Endospore formation

- ✓ under unfavourable conditions.
- Mostly found in Bacillus. ✓



Mature endospore gives rise to one new bacteria after germination.

Reproduction in Bacteria

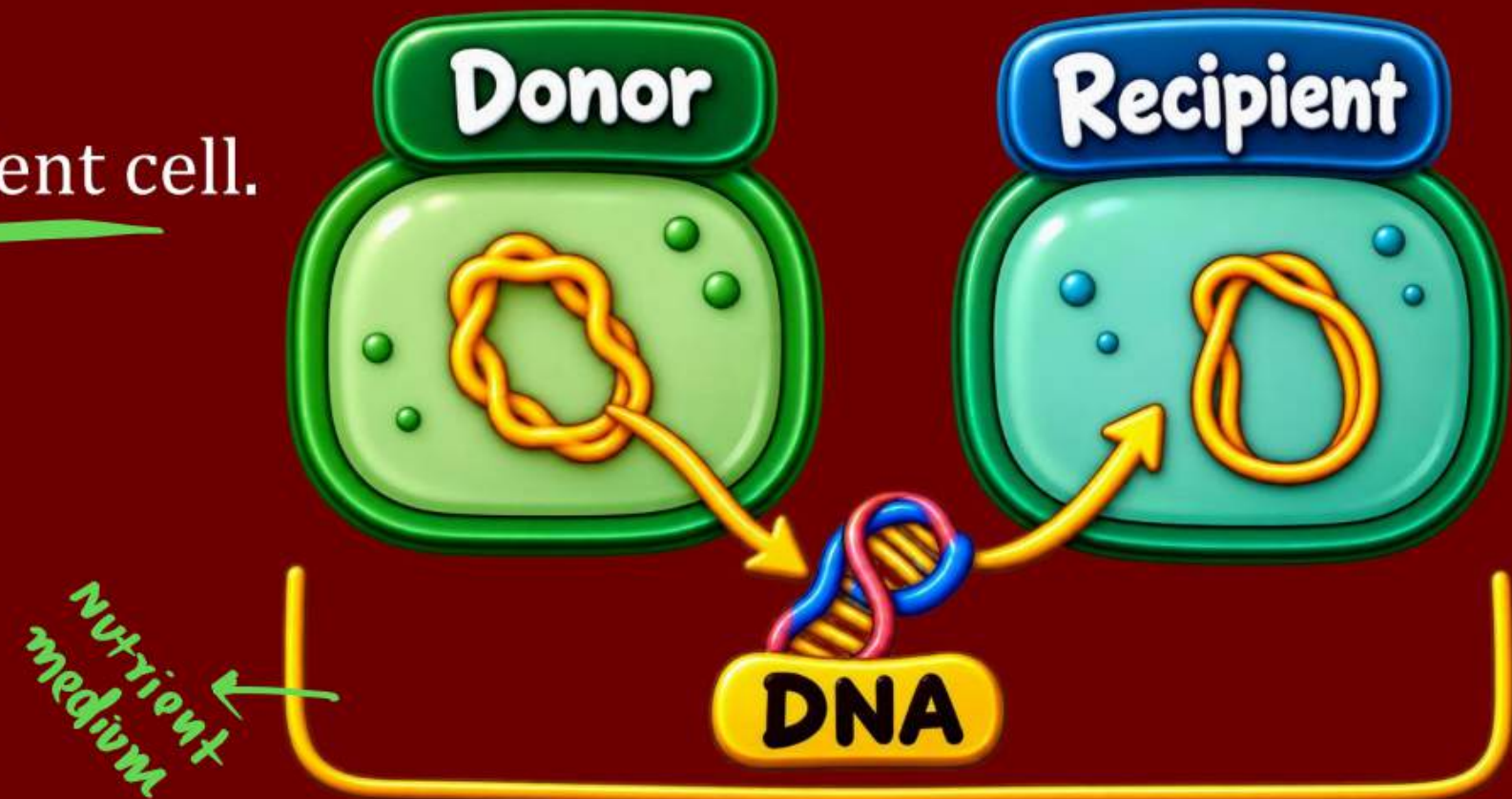
3. Genetic Recombination (Sort of Sexual reproduction)

Following types.

✓ a. Transformation:

- Donor cell releases a piece of DNA & it is actively taken up by recipient cell from medium.
- No contact b/w Donor & recipient cell.

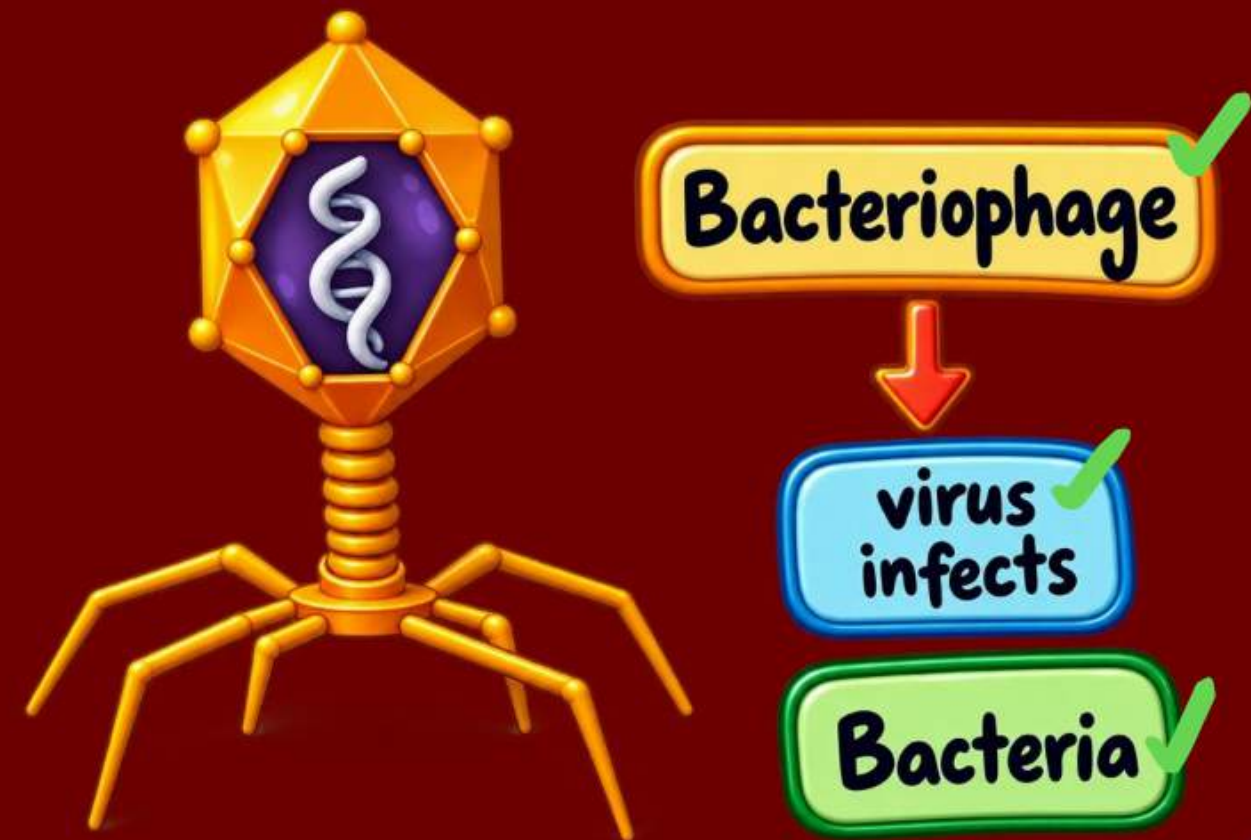
Ex. Streptococcus pneumoniae.



b) Transduction

- DNA transfer from one bacterium to another takes place by a virus known as bacteriophage

Ex. Salmonella typhii



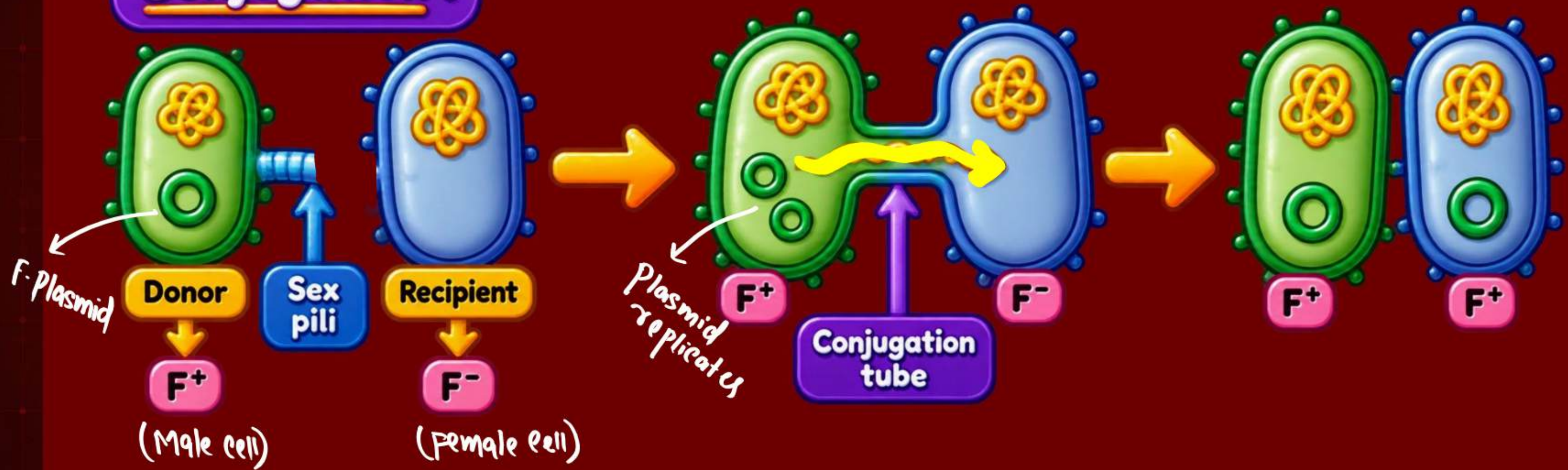
c) Conjugation



- Two cells involved
 - ✓ Donor cell (F^+) (possess F-plasmid)
 - ✓ Recipient cell (F^-) (lacks F-plasmid)
- Donor cell is attached to recipient cell by Sex Pilli.
- Sex Pilli functions as Conjugation tube.
- F-plasmid now replicates & the replica moves to F^- cell via Conjugation tube.
- Both cells get separated.
- Due to transfer of F-plasmid, F^- bacteria now becomes F^+ bacteria.

Example: E. Coli.

Conjugation :



CLASSIFICATION OF BACTERIA

Archaeobacteria

- **Habitat** : Extreme Environmental conditions.
- **Cell wall** : Composed of pseudomurein.
- **03 TYPES** :

a Methanogens

- Marshy habitat.
- ✓ Present in gut of ruminant animals.
- ✓ Produce biogas (CH_4) from dung of animals.
- Ex. Methanobacterium



b Halophiles

- ✓ Present in high salt areas.
- ✓ Ex. Halobacterium



c Thermoacidophiles

- ✓ Found in :
hot H_2O springs ($80-100^\circ\text{C}$)
- low pH = 2 (acidic)
- ✓ Ex. Thermus aquaticus



Eubacteria

Eubacteria include

- Cyanobacteria (BGA) ✓
- Mycoplasma ✓

Cyanobacteria



Habitat: Fresh H₂O, Marine H₂O, Terristrial.

Body form:

↓
Salt Conc.
↓↓

↓
Salt Conc.
↑↑

✓ • Unicellular



✓ • Filamentous



✓ • Colonial.



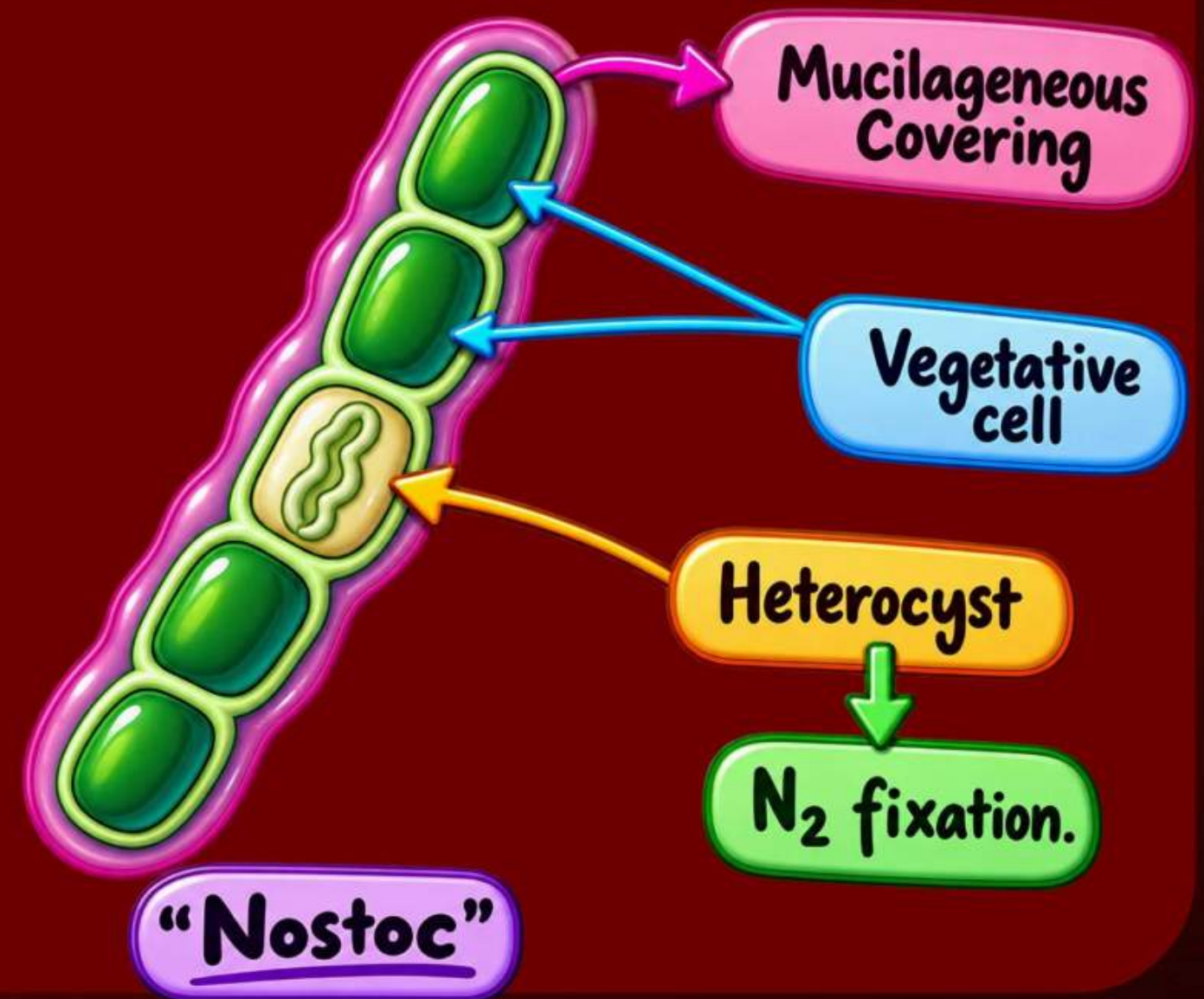
Cell wall: Made of peptidoglycan.

- PM — possess invaginations called as chromatophores ^{which} c bear pigments.
- Mesosomes — ant.

Cyanobacteria

- Some cyanobacteria fix atmospheric N_2 in specialised cells kid as heterocyst.

Ex. Nostoc, Anabaena.



Water Bloom or Algal Bloom

- Excessive growth of algae in aquatic ecosystems.
- BGA (Cyanobacteria) grows rapidly in water and secrete toxic substances which causes death of aquatic animals.
- It also gives toxicity and bad smell to water.

Mycoplasma



- AKA PPLO (Pleuro Pneumonia Like Organisms).
- discovered in pleural fluid of cattle suffering from pleuropneumonia disease.
- Smallest living cell. (0.1 – 0.3 μm)
- Cell wall Absent.
- No regular shape (i.e why kld as 'Joker of microbial world.')
- Resistant to antibiotic- Pencillin ($\because$ acts on cell wall).
- Osmotrophic mode of Nutrition (absorption of nutrients by osmosis).
- Double stranded linear DNA — Present.
- No histone proteins.

} Pleiomorphic
↓
No regular shape.



Mycoplasma



- Many mycoplasma are pathogenic in plants & animals.
 - **Animal diseases —**
 - Pneumonia ✓
 - Reproductive disorders. ✓
 - **Plant diseases —**
 - Little leaf disease of Brinjal. ✓
 - Papaya bunchy top. ✓

Kingdom Protista



- **INTRODUCTION:**

- ✓ Introduced by Ernest Haeckel.
- ✓ it includes Unicellular Eukaryotes.
- ✓ Membrane bound organelles — Present.
- ✓ Nucleus — Present.

- **Habitat:** primarily Aquatic.

- Boundries of this Kingdom are not well defined.

Reason: Some members shares characters with other members of other group.

Classification (Protista) :

1. Chrysophytes



2. Dinoflagellate



3. Euglenoids



4. Slime moulds



5. Protozoans



Photosynthesis
(plants like)



Saprophytic
(like fungi)



Heterotrophic
(like animals)



Chrysophytes



- Includes — Diatoms & Desmids.
 - ✓ (little knowledge is known about desmids.)

Habitat: Fresh H₂O , Marine H₂O

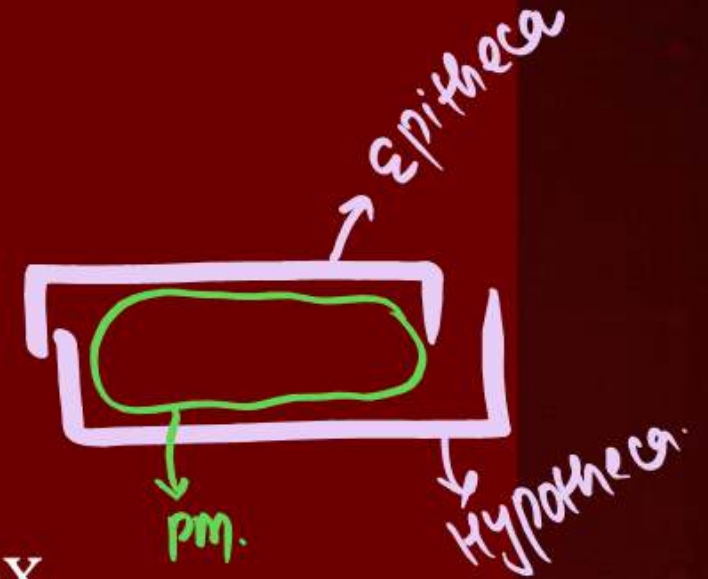
Flagella: Absent (Non-motile)

Cell wall: Cellulose + Silica

- Possess two thin overlapping shells that fit together like soap box.
- Cell wall — indestructible because of presence of Silica.
- When diatoms die, they accumulate at the bottom of H₂O bodies and thus form Diatomaceous Earth / Diatomite.

Pigment: Chl-a, Chl-c, fucoxanthin.

- Perform photosynthesis.
- Diatoms — Chief producers in oceans.



Diatomaceous Earth

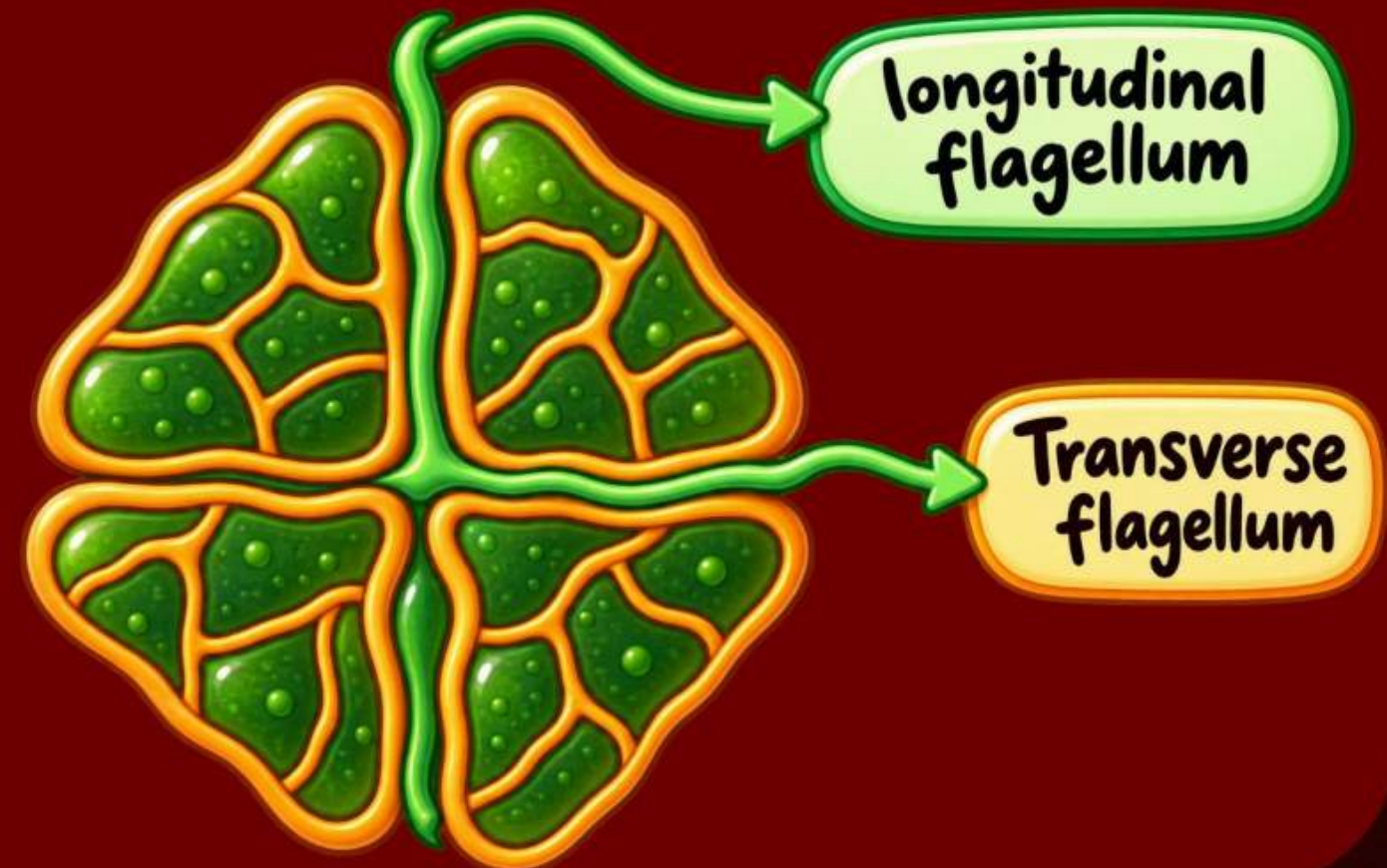
- Diatomaceous Earth being highly porous and gritty in nature, it is used in:
 - a) filtration of oils & syrups.
 - b) polishing of metals.

Dinoflagellates

Habitat: Marine.

Cell wall: Cellulose + Pectin.

- Its outer surface possess stiff cellulosic plates.
- Covering is seen as 'armoured' i.e, why these are AKA armoured algae.



Dinoflagellates



Flagella: Two in no. ✓

✓ Transverse ✓

✓ Longitudinal ✓

- i.e, why kid as dinoflagellates

Pigments: Chl-a, Chl-c, Xanthophyll.

- Perform photosynthesis.
- Depending on main pigment +nt, Dinoflagellates appear — yellow, Green,
Brown, Blue, Red.

Ex: Red dinoflagellates — phycoerythrin is +nt in abundance.

Dinoflagellates



- Most common mode of reproduction is Binary fission.

Red dinoflagellates like Gonyaulax

- Rapid multiplication (by Binary fission)
- Sea appear Red. (RED TIDES)
- These large number of Red dinoflagellates release toxins & may even kill other marine animals eg. Fishes.

Euglenoids



Habitat: Fresh H₂O.

- Stagnant H₂O.

Cell wall: ABSENT.

- Outside PM is thick protein rich layer — PELLICLE. → **Flexible**

• Flagella:

- Two in no. ✓

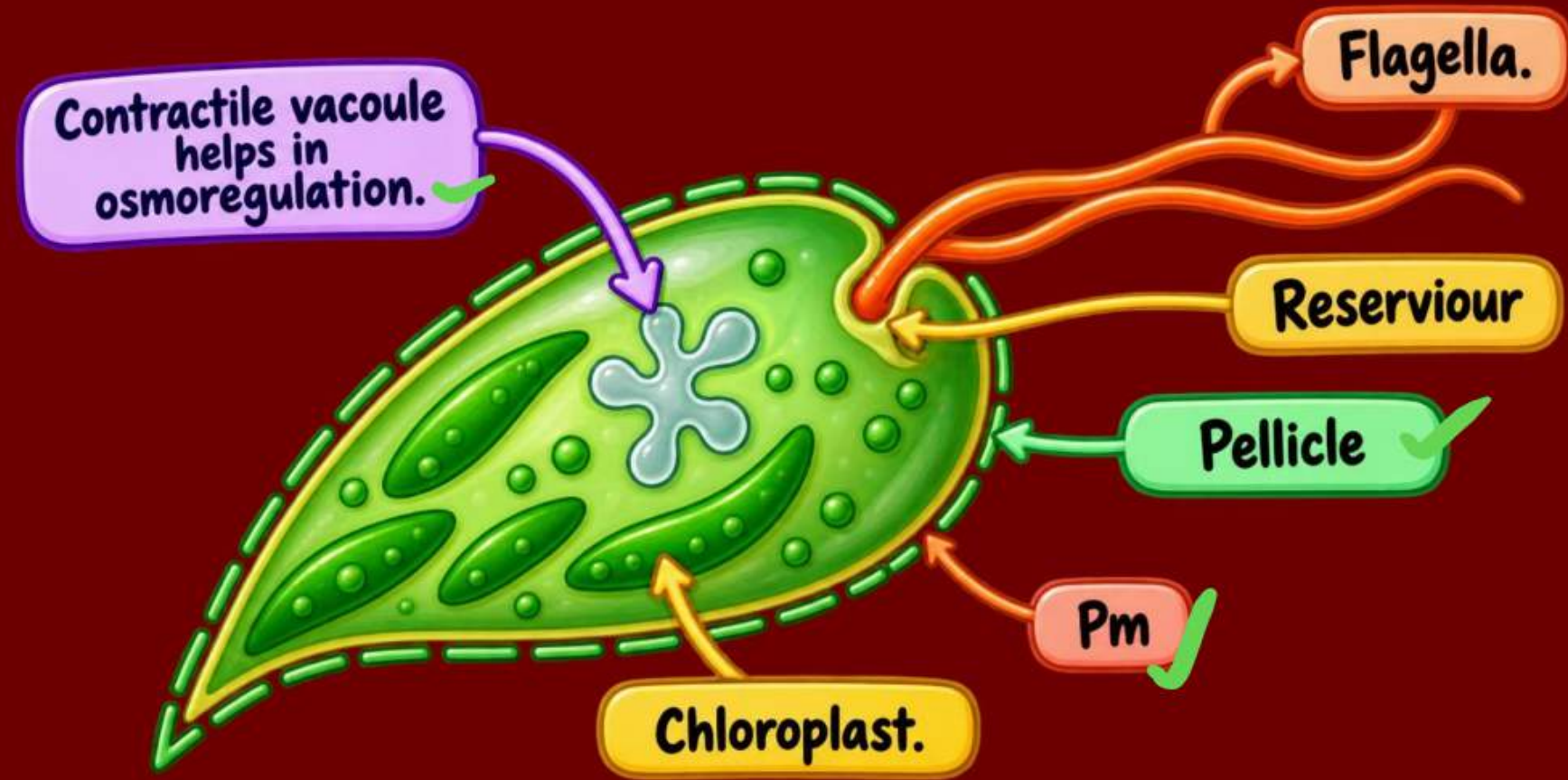
- long (01)

- short (01)

- **Pigments:** Chl-a, Chl-b, Xanthophyll.

→ higher plant like character.

Euglenoids



Nutrition:

- Mixotrophic
 - ✓ Photosynthetic in +nse of Light.
 - ✓ In -nse of Sunlight, they act as heterotrophs by predating on other smaller organisms.
- Euglena — connecting link b/w animal Kingdom and Plant Kingdom.

Slime Moulds



- (FALSE FUNGI) ✓

Habitat: Terristrial.

Cell wall: ABSENT.

Chlorophyll: ABSENT.

Nutrition: Saprophytic

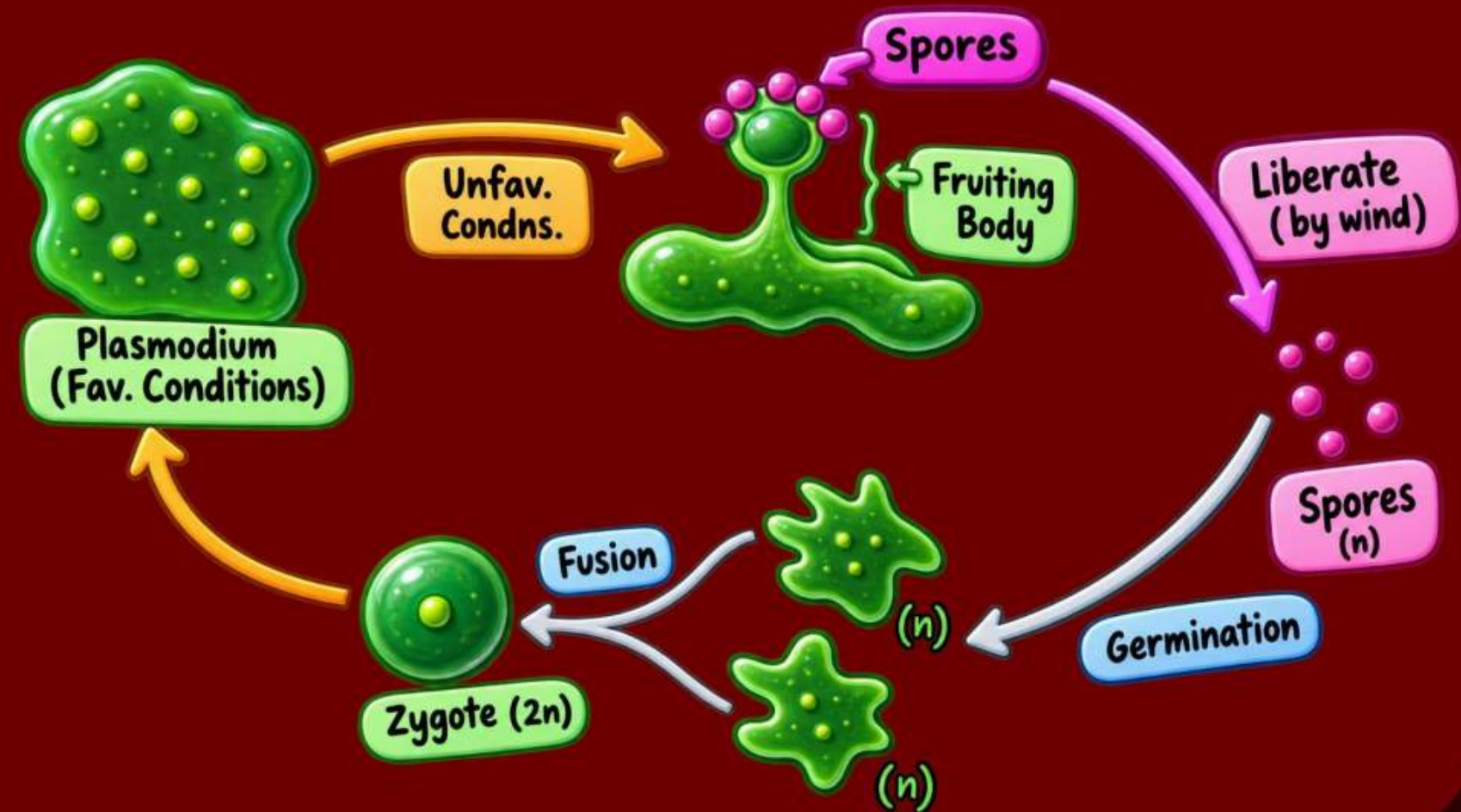
Favourable Conditions: Slime moulds form an aggregation kd as 'Plasmodium'.
grows & spread over several feet.

Unfavourable condition — Plasmodium diffrentiates and form fruiting bodies,
bears spores at their tips.

Slime Moulds

SPORES →

- ✓ True cell wall + nt.
- ✓ Extremely resistant.
- ✓ Survive for many years.
- ✓ Dispersed by air currents.



Protozoans



- Primitive relatives of animals.
- All are heterotrophs (Predators or Parasites)

Cell wall: ABSENT.

04 major groups: (on the basis of locomotory organ).

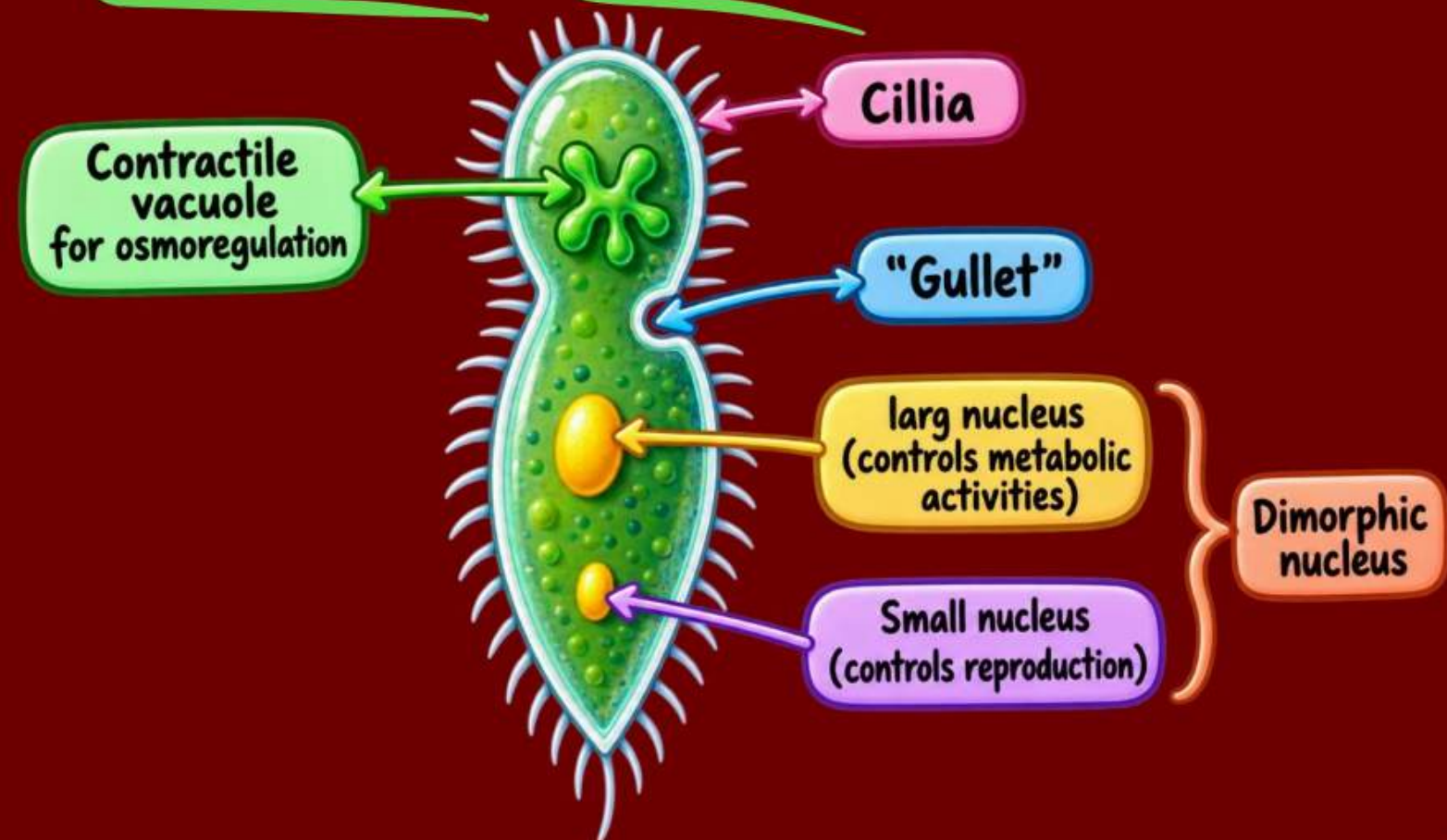
1. Ciliated protozoans. ✓
2. Amoeboid protozoans. ✓
3. Flagellated protozoans. ✓
4. Sporozoans. ✓

Four Types of Protozoans — Ciliated Protozoans

1. Ciliated protozoans:

- Aquatic (few are parasitic).
- Locomotory organ — Cilia (thousands in no.)
- Cilia — also helps in ingestion of food through 'Gullet'

Ex. Paramecium.

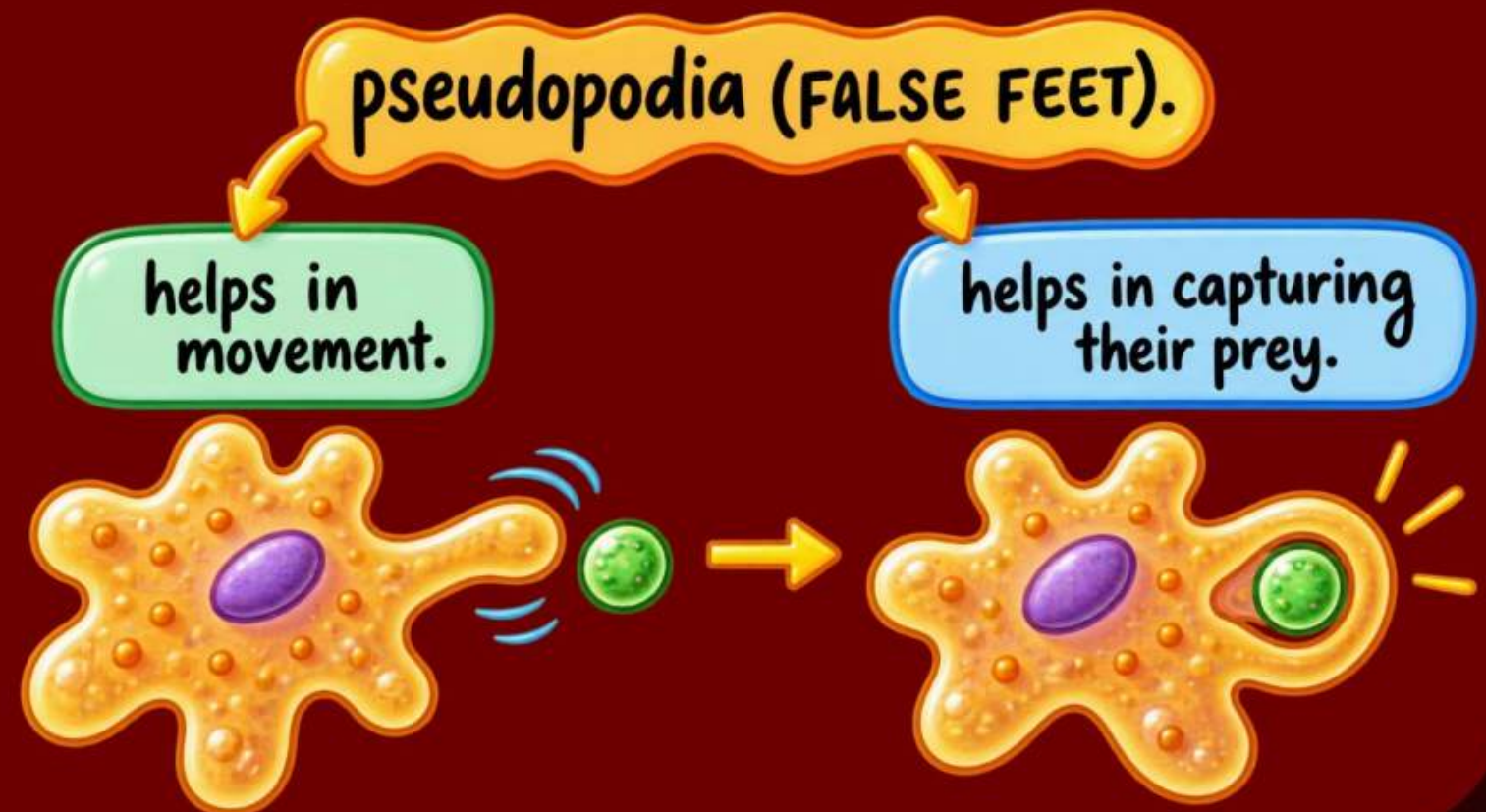


Four Types of Protozoans —

2. Amoeboid protozoans:

Habitat:

- Fresh H₂O, marine, moist Soil.
- Marine forms possess — Silica shell on their surface.
- Free living (Ex. Amoeba)
- Parasitic (Ex. Entamoeba)



Four Types of Protozoans —

3. Flagellated protozoans:

- ✓ Free living (Aquatic)
- ✓ Parasitic.
- ✓ Locomotory organ is Flagella.

Ex. Trypanosoma (parasitic) — Causes 'Sleeping Sickness' disease.

4. Sporozoans:

- ✓ All are Endoparasites (inside tissue).
- ✓ Locomotory organ — ABSENT.
- ✓ Their life cycle possess — infectious spore like stage.

Ex. Plasmodium (malarial parasite) — Causes malaria in humans.

Kingdom Fungi



Introduction

- ✓ Study of fungi = Mycology
- ✓ Achlorophyllous (chlorophyll -nt)
- ✓ Heterotrophic ✓
- ✓ Saprophytic (obtains food from dead or decaying organic matter)
- ✓ Parasitic ✓
- ✓ Thalloid (Not differentiated into Root, Stem & leaves)

Kingdom Fungi



Habitat:

- Everywhere
- Grows in humid or high temp. areas

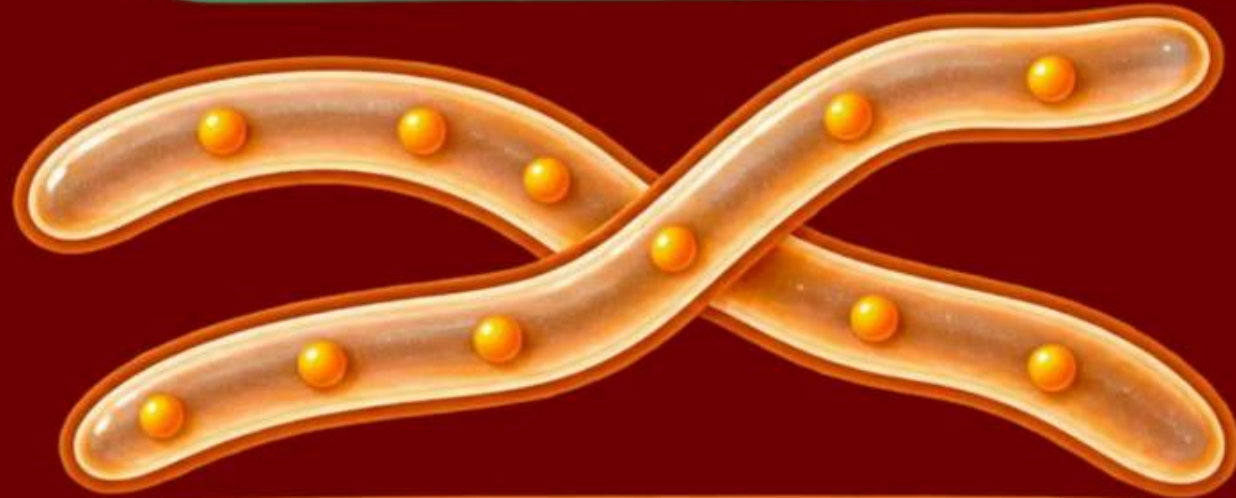
Body Forms — Hyphae & Mycelium

- Multicellular ✓
- Exception: Yeast (unicellular)
- Multicellular forms possess network of hyphae, kid as Mycellium.
- Hyphae: long, slender, thread like structures
- Hyphae combine to form Mycellium



Mycellium (02 types)

Aseptate mycellium



- Septa - nt.
- Coenocytic (multinucleated)

Septate mycellium



- Septa/wall b/w nuclei is +nt.

Nutrition:

- Heterotrophic
- Saprophytic (max.)
- Parasitic

Cell Wall:

- Present (+nt)
- Made of Chitin

Lichen:

- Symbiotic association b/w fungi & algae

Micorrhiza:

- Fungi associated with roots of higher plants (Pinus)

Reproduction

- 1) Vegetative reproduction
- 2) Asexual reproduction ✓
- 3) Sexual reproduction ✓



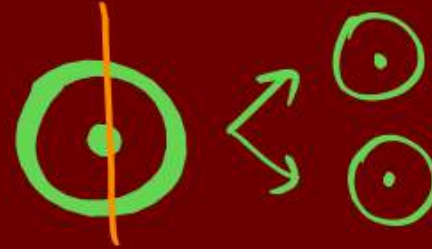
Reproduction



a. Fragmentation

- Fungal filament breaks into small pieces due to any reason and forms new fungi
- Most common.

b. Fission



- Fungal cell divides into two parts along with nucleus.

Ex. Schizosaccharomyces (Fission yeast)

c. Budding



- Bud is formed which gets detached and functions as young fungi.

Ex. Saccharomyces (Budding yeast)

2. Asexual Reproduction

- By asexual spores.
- Types of asexual spores.

i) **Sporangiospores** ✓

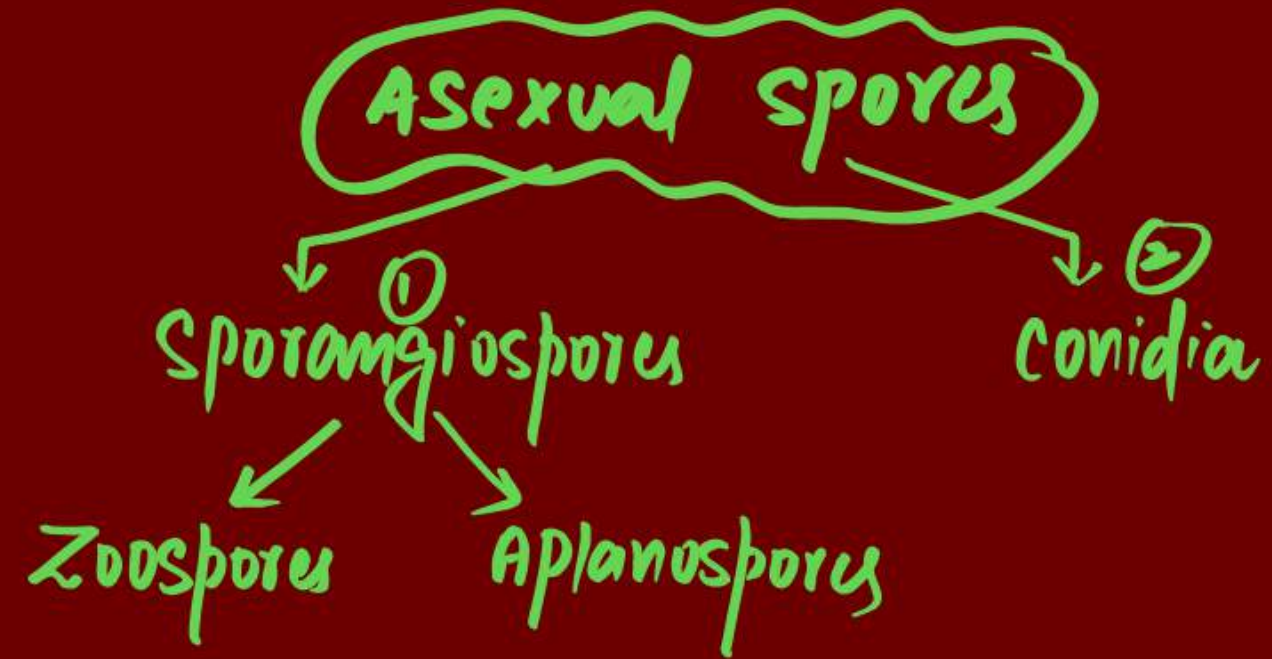
- Formed inside sporangium.
- Endogenously formed (inside sporangium).

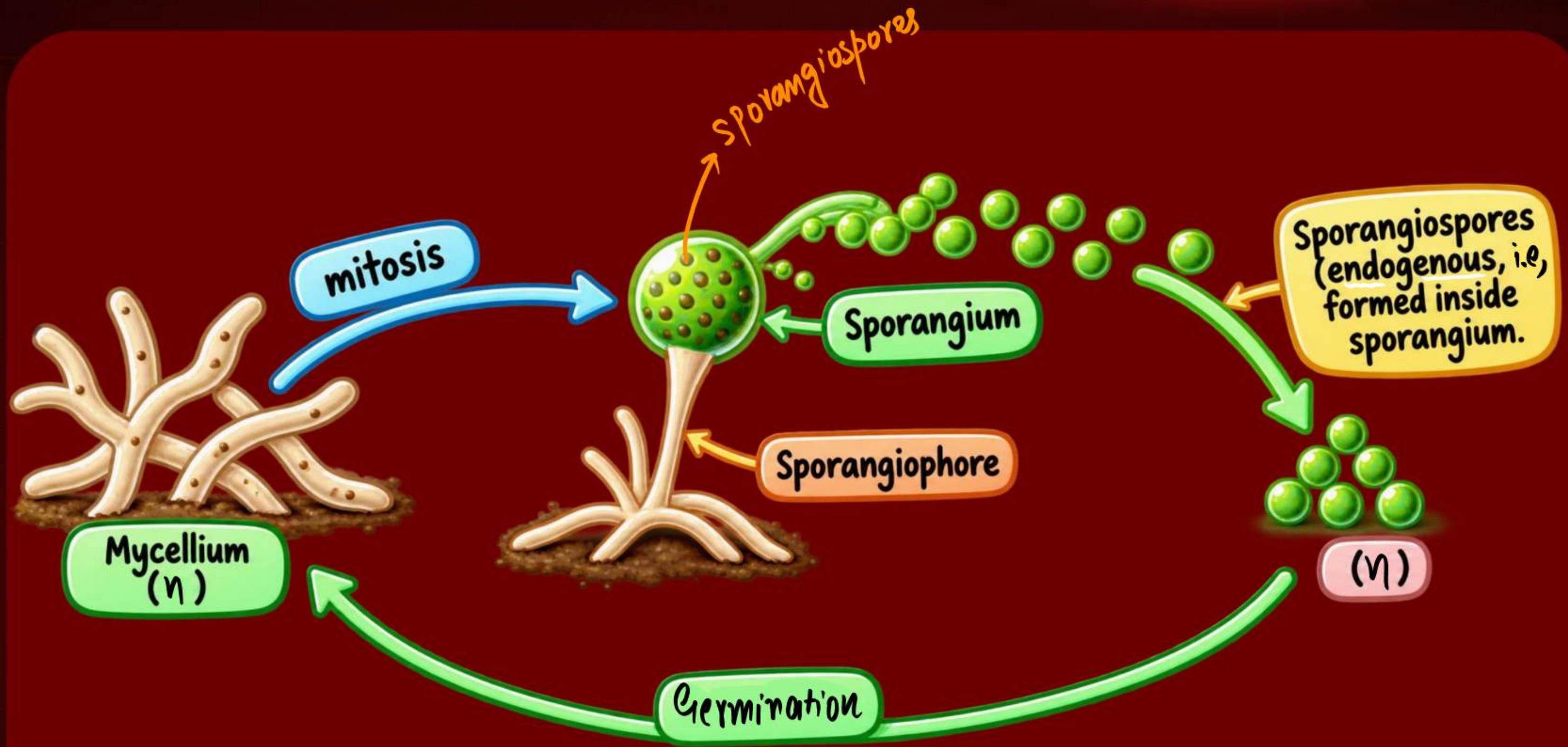
Two types:

a) **Zoospores**

- Flagellated (motile)
- Naked (no cell wall)

Ex. Aquatic members of Phycomycetes.





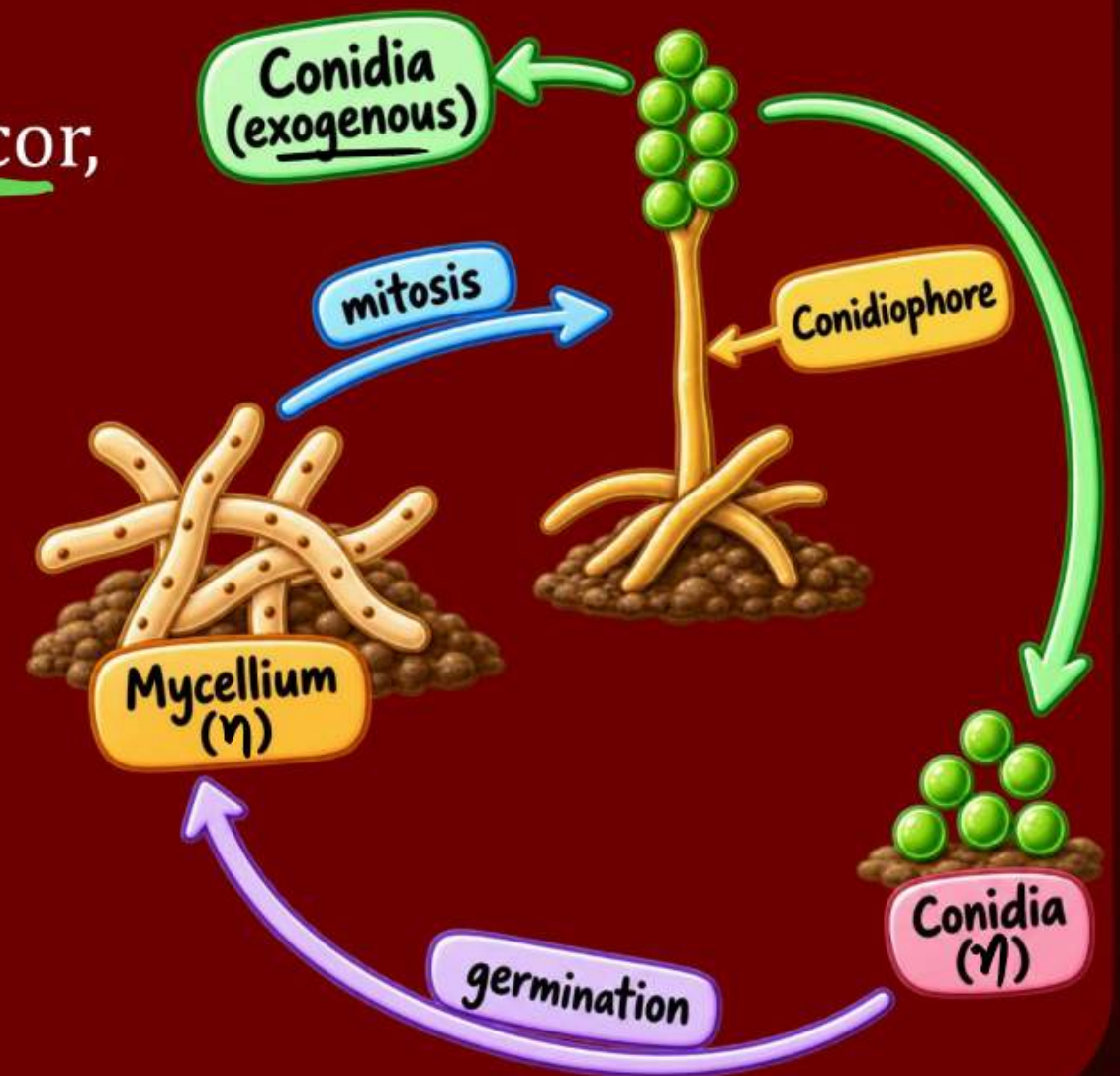
2. Asexual Reproduction

b) Aplanospores

Non-flagellated (non-motile)

Thin walled.

Ex. Terristrial members of Phycomycetes like Mucor,
Rhizopus.



2. Asexual Reproduction

ii) Conidia

Most common asexual spore

Formed at tip of conidiophore.

Exogenously formed.

Ex. Ascomycetes, Deuteromycetes.

NOTE: Asexual spores — generally absent in Basidiomycetes.

Ascomycetes
Phycomycetes

Basidiomycetes → No asexual
Spores.

Deuteromycetes

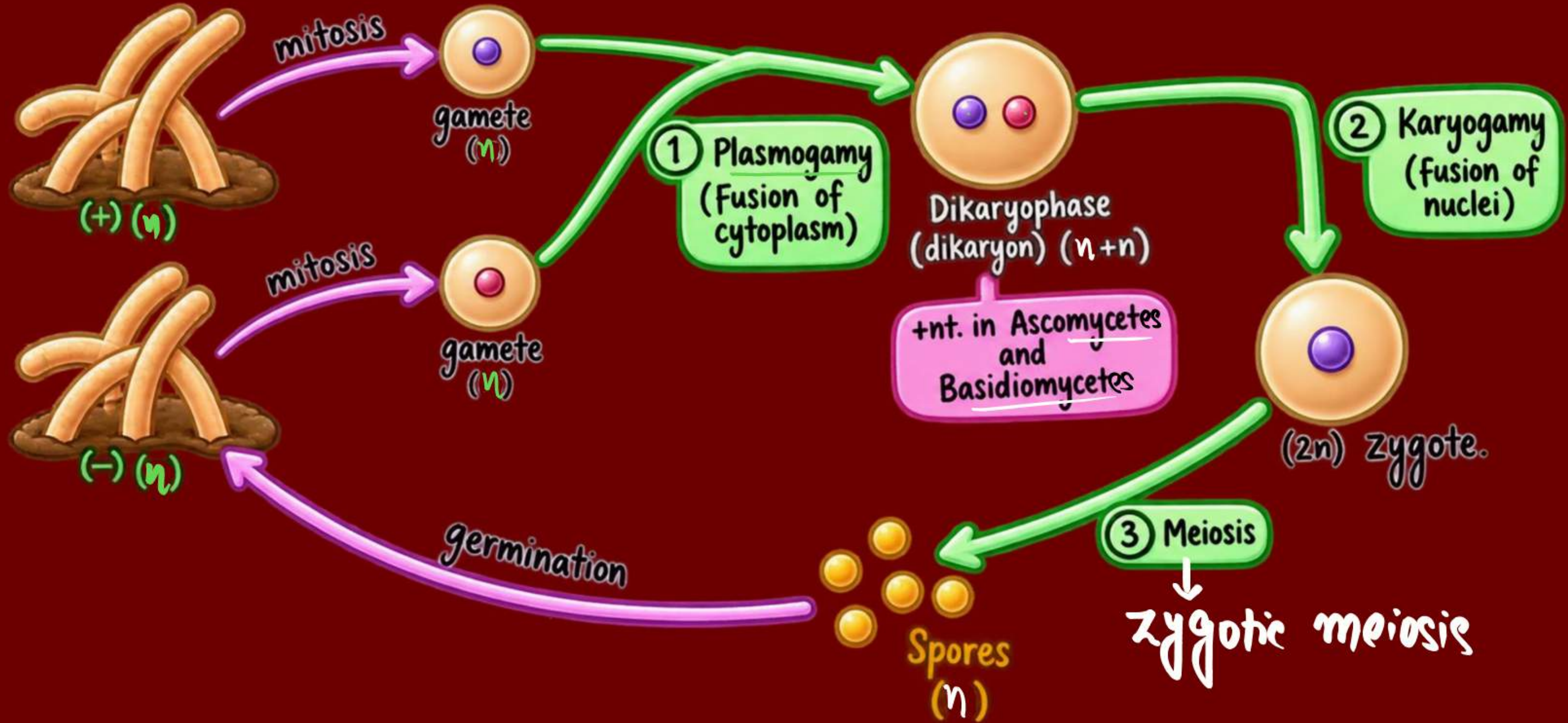
3. Sexual reproduction

involves

① Plasmogamy

→ ② Karyogamy

→ ③ Zygotic meiosis



Depending upon:

1. Morphology of mycelium ✓
2. Mode of spore formation ✓
3. Mode of fruiting body formation ✓

Fungi is divided into 04 classes:

- (1) Phycomycetes ✓
- (2) Ascomycetes ✓
- (3) Basidiomycetes ✓
- (4) Deuteromycetes ✓

Phycomycetes (Lower Fungi)

Common name: Algal fungi

Habitat and nutrition: Aquatic, decaying wood, moist & damp places or as obligate parasites on plants.

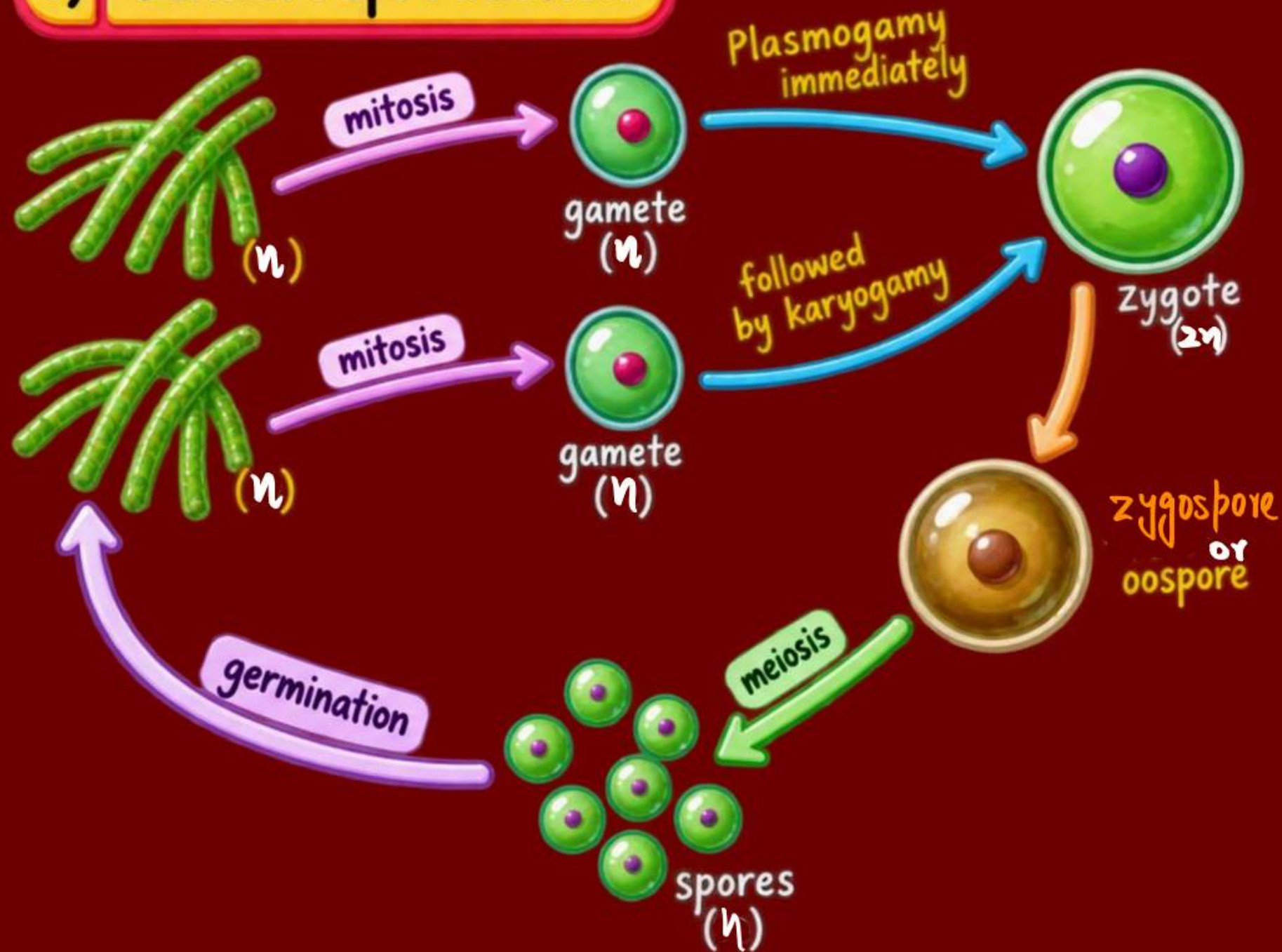
Mycellium: Aseptate and coenocytic (multinucleated)

Reproduction:

- a) Vegetative reproduction: Fragmentation
- b) Asexual reproduction: Zoospore or Aplanospore

PHYCOPHYCTES

c) Sexual reproduction:



Sexual spore:

Zygospore
or
oospore.

Fruiting body:

ABSENT.

Gametes:

PRESENT

Morphologically
similar

Morphological
dissimilar

Phycomycetes (Lower Fungi)

Dikaryophase: Absent

Examples:

- Mucor (Dung mould)
- Rhizopus (Bread mould)
- Albugo candida

Ascomycetes



Common name: Sac fungi

Habitat and nutrition:

On decaying organic matter, decomposers

Coprophilous (grows on dung), Saprophytic, Parasitic

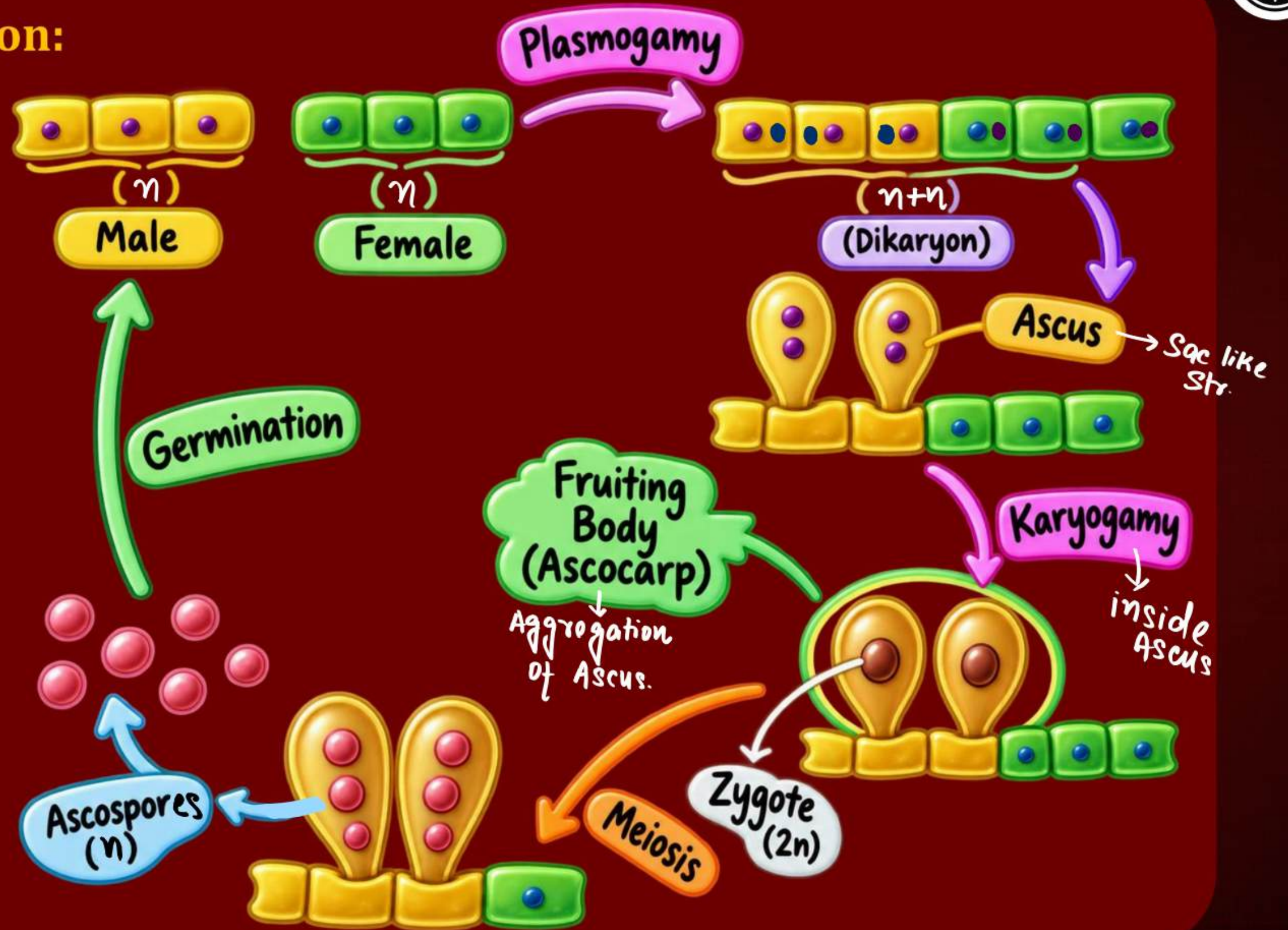
Mycellium: Branched and septate

Reproduction:

a) Vegetative reproduction: Fragmentation, Budding, Fission

b) Asexual reproduction: Conidia

c) Sexual reproduction:



Sexual spore:

Ascospores — formed endogenously inside Ascus

Fruiting body:

- present.
- Ascocarp

Gametes: $+nt$.

Dikaryophase: present.

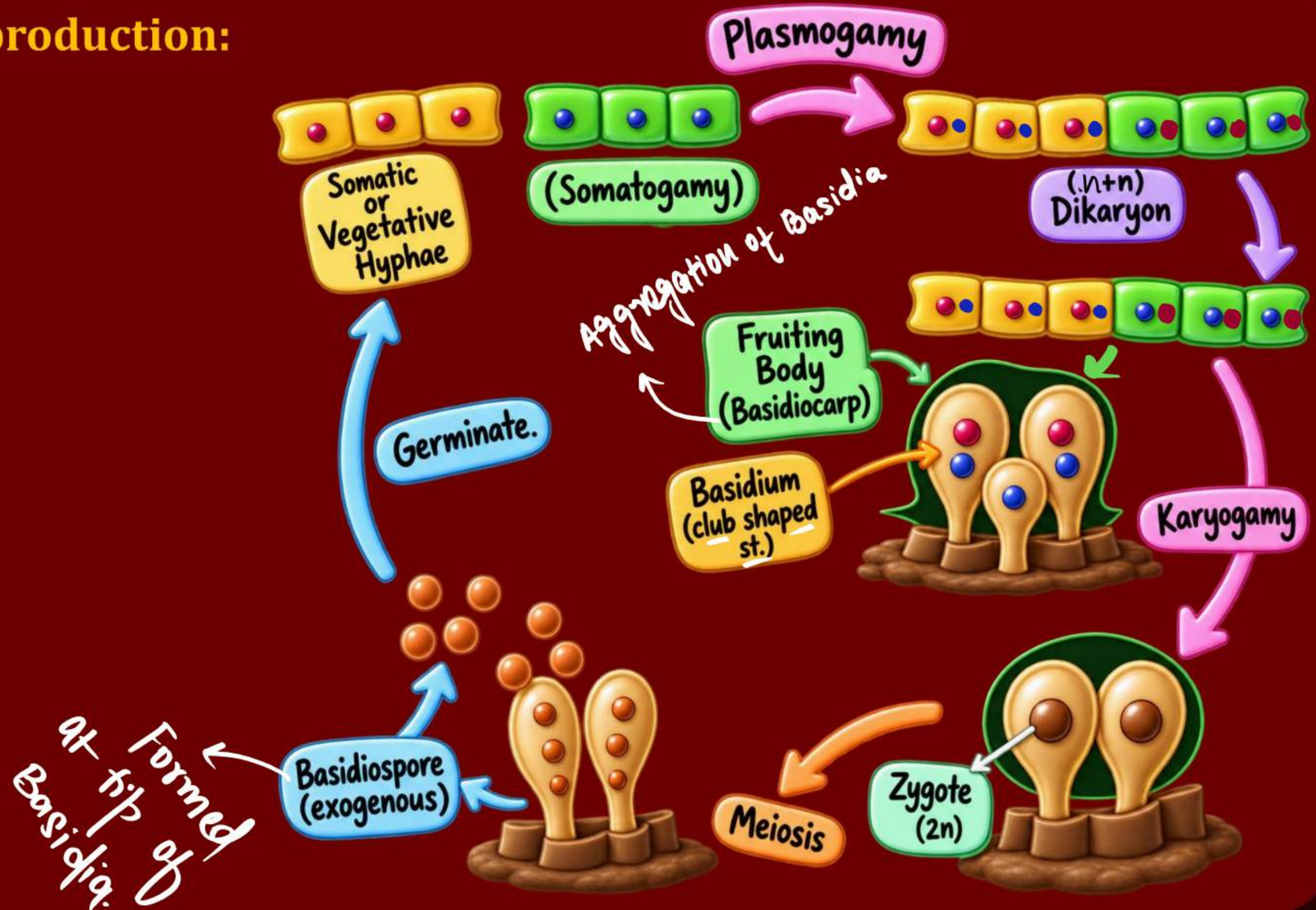
Examples: Yeast, Pencillium, Aspergillus, Claviceps, Morels & trufles, Neurospora

Basidiomycetes



- **Common name:** Club fungi
- **Habitat and nutrition:**
 - ✓ Soil, wooden logs, tree stumps
 - ✓ Parasites on living plant bodies (eg. Rusts, Smuts)
- **Mycellium:** Branched and septate
- **Reproduction:**
 - a) Vegetative reproduction: Fragmentation
 - b) Asexual reproduction: Asexual spores - nt

c) Sexual reproduction:



Sexual spore: $+nt$

Basidiospore — formed at tip of Basidium (exogenous)

Fruiting body: $+nt$ Basidiocarp

Gametes: $-nt$ (Somatogamy)

Dikaryophase: $+nt$

Examples: Agaricus, Bracket fungi, Puffballs, Ustilago (Smut in wheat), Puccinia (Rust fungi), Toadstools (Poisonous)

Deuteromycetes

- **Common name:** imperfect fungi
- **Habitat and nutrition:**
 - ✓ Grows on litter
 - ✓ Maximum are decomposers (help in mineral cycling)
 - ✓ Some — Saprophytic / parasitic
- **Mycellium:**
 - ✓ Septate and Branched
- **Reproduction:**
 - a) Vegetative reproduction: Fragmentation
 - b) Asexual reproduction: Conidia

Deuteromycetes



c) Sexual reproduction: ABSENT

NOTE: When sexual forms of these fungi were discovered, they were moved into the class they rightly belong to.

Sexual spore: ABSENT ✓

Fruiting body: ABSENT ✓

Gametes: ABSENT ✓

Dikaryophase: ABSENT ✓

Examples: Alternaria, Trichoderma, Colletotrichum

Viruses, Viroids, Prions and Lichens

Virus:

Study = Virology

Acellular (Exception to Cell theory)

Requires host to complete its life cycle

Viruses - Obligate intracellular parasites

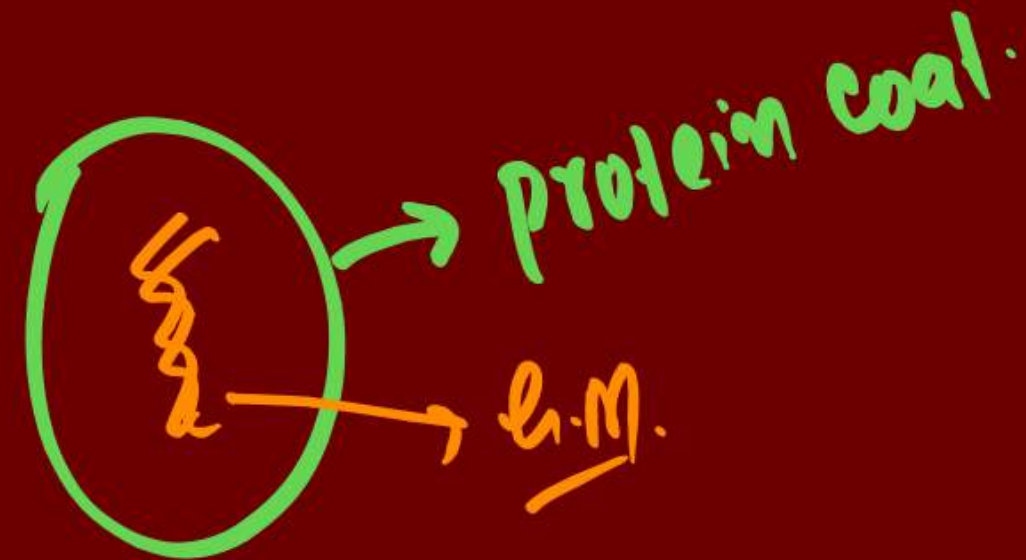
Contain:

Protein Coat ✓

Nucleic Acid (DNA/RNA) ✓

Viruses are living → inside living cell

Viruses are dead → outside host/living cell



STRUCTURE OF VIRUS

1 Protein Coat

forms external protective covering.

Made up of small subunit kld as **capsomers**.



capsomer

2 Nucleic acid

genetic material

either DNA | RNA.

ds DNA

ss DNA

ss RNA

ds RNA

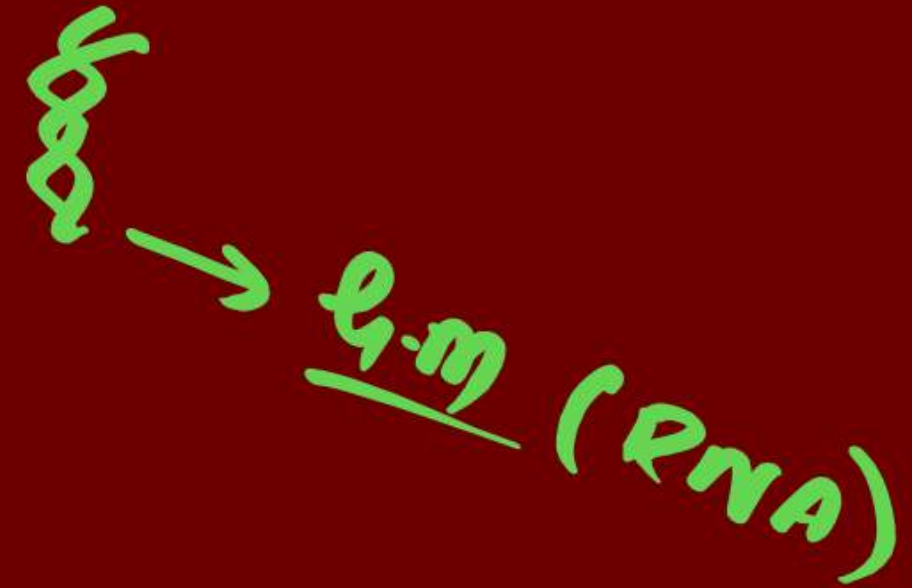
max. ✓

Viroids and Prions



Viroids:

- Discovered by T.O Diener (1971).
- Infectious particle without protein coat.
- Infectious particle is RNA.
- Smaller than Viruses.
- Causes PSTD (Potato Spindle Tuber Disease).

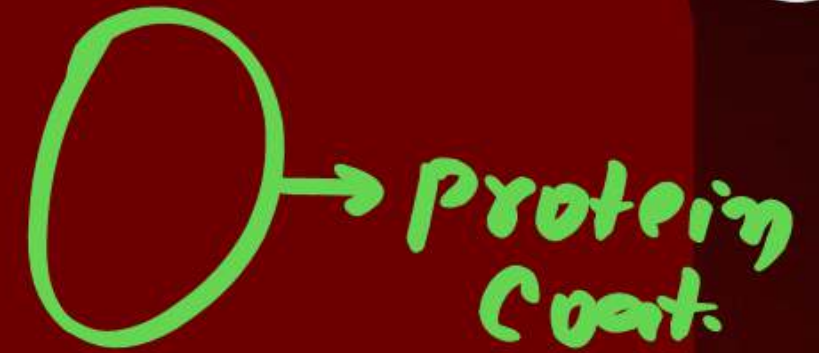


Viroids and Prions



Prions:

- Infectious protein particles without genetic material.
- Causes infection by changing the folding of other proteins.
- Size equal to Viruses.
- Cause disease in cattle — BSE (Bovine Spongiform Encephalopathy)
- AKA Mad Cow disease.
- Also causes CJD disease in humans (Creutzfeldt-Jacob disease) →
Analogous to BSE.



Lichens:

Symbiotic association b/w algae and fungi.

Two partners (c benefit to each other)

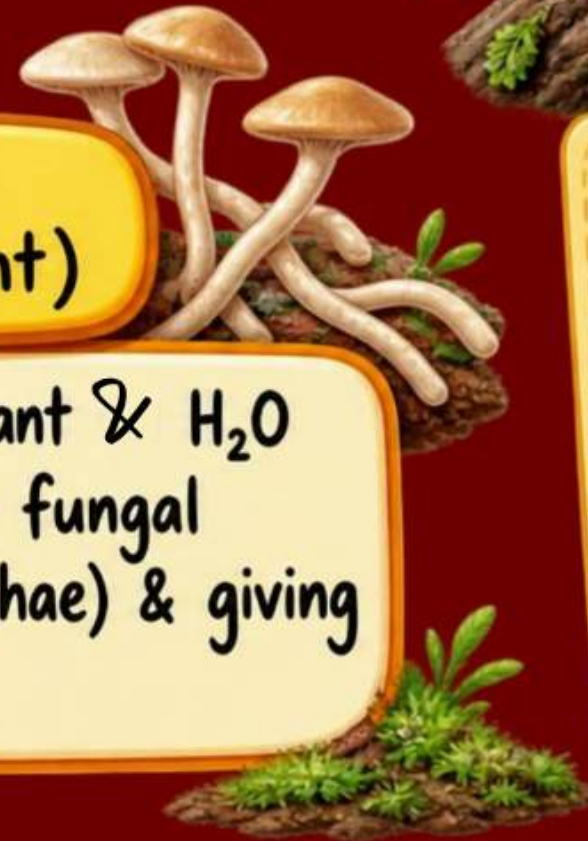
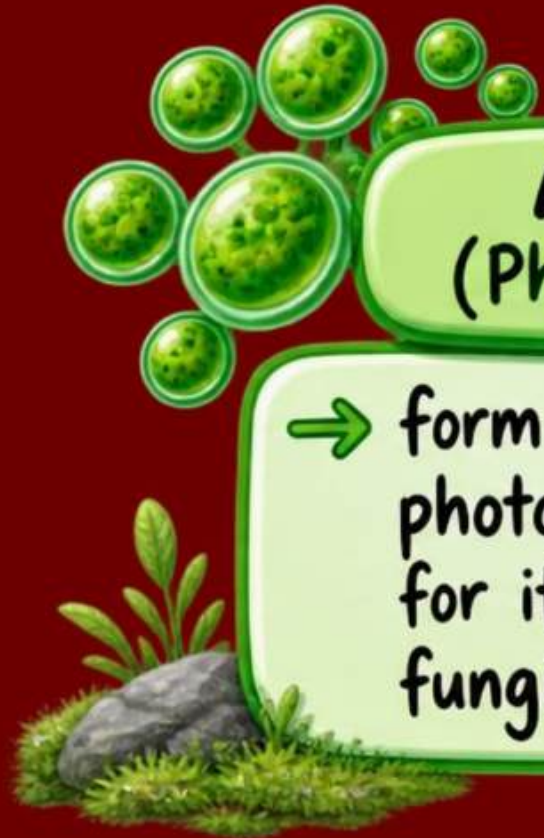
Algae
(Phycobiont)

→ form food by photosynthesis for itself & also for fungi.

Fungi
(Mycobiont)

→ absorbs nutrient & H_2O from soil by fungal filament (hyphae) & giving it to algae.

Note: ✓
Lichens
↓
very good pollution indicators.



Most IMPORTANT Questions for JKBOSE 2026

- 1) Discuss artifitial and natural classification? } *Plant Kingdom*
- 2) Two-kingdom classification & 5-Kingdom classification?
- 3) **Discuss:**
 - a) Nutrition in bacteria b) Reproduction in bacteria.
- 4) Characteristic features of – Monera, Protista, Euglenoids, Fungi?
- 5) Discuss sexual reproduction in fungi?
- 6) Discuss – Viruses, Viroids, Lichens?



Thank
You